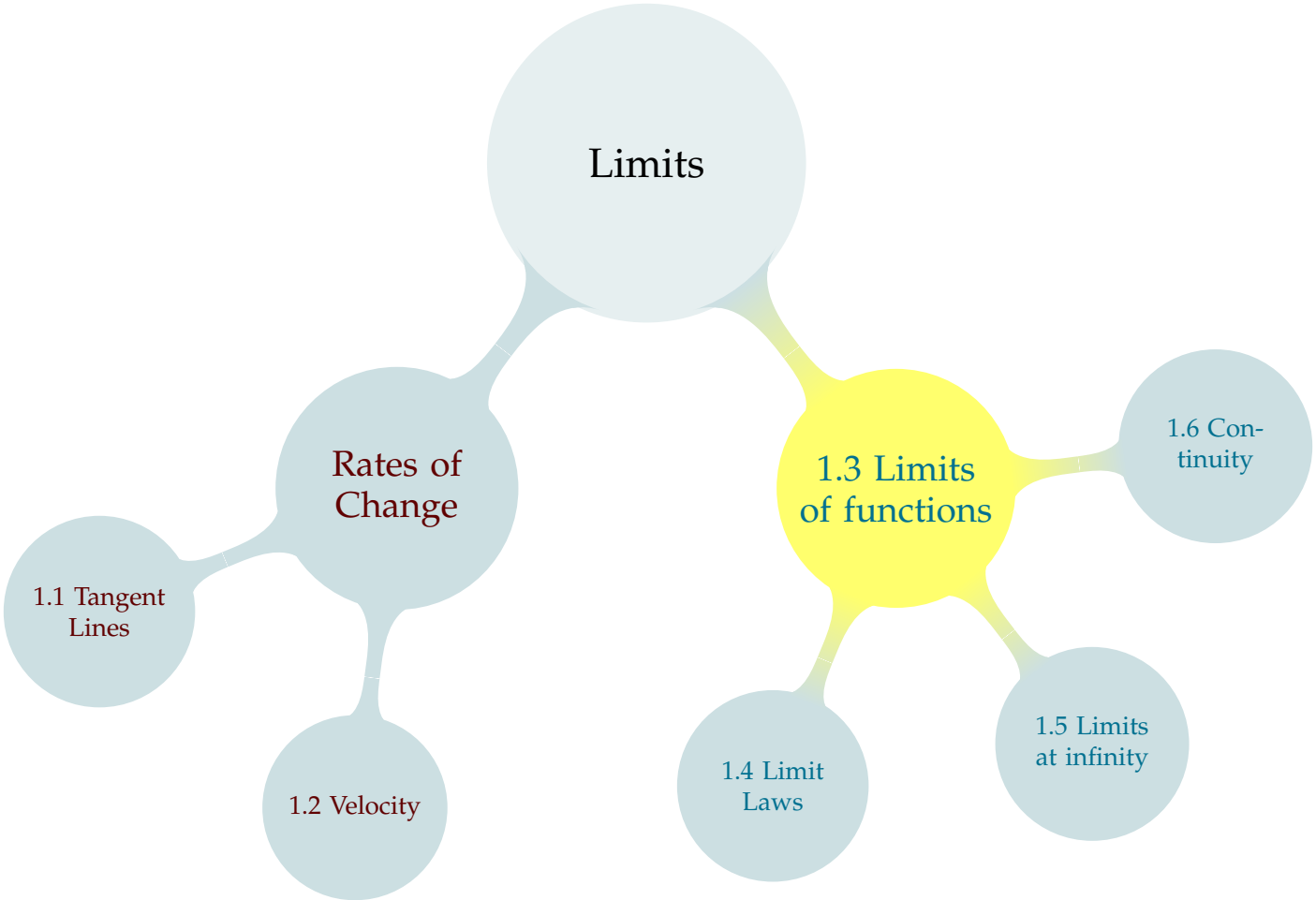
Table of Contents



Notation 1.3.1 and Definition 1.3.3

$$\lim_{x \to a} f(x) = L$$

where a and L are real numbers

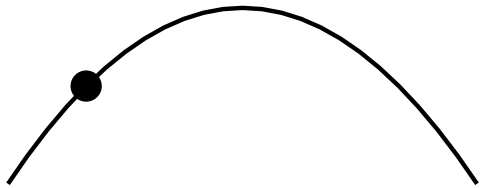
We read the above as "the limit as x goes to a of $f(x)$ is L."

Its meaning is: as x gets very close to (but not equal to) a, $f(x)$ gets very close to L.

Finding Slopes of Tangent Lines



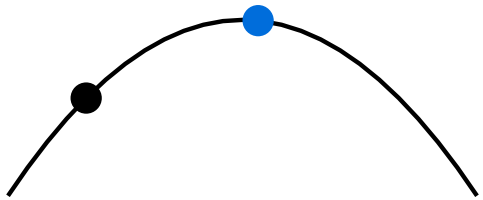
We NEED limits to find slopes of tangent lines.



Finding Slopes of Tangent Lines



We NEED limits to find slopes of tangent lines.

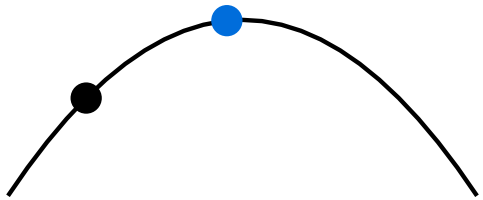


Slope of secant line: $\frac{\Delta y}{\Delta x}$, $\Delta x \neq 0$.

Finding Slopes of Tangent Lines



We NEED limits to find slopes of tangent lines.

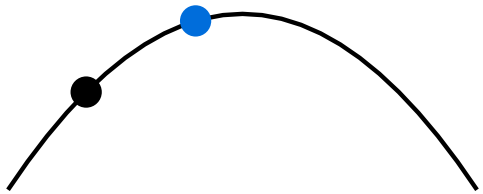


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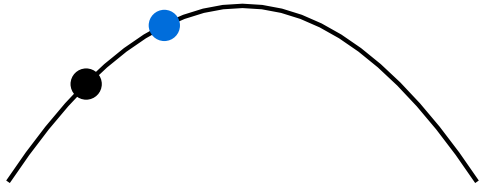


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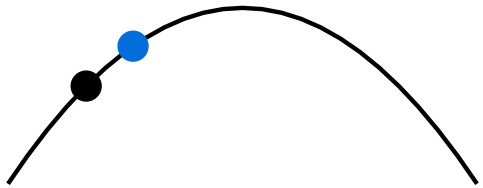


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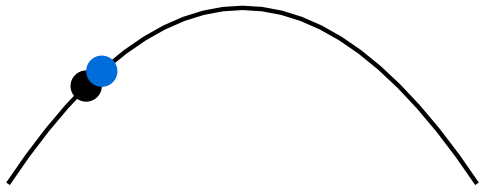


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Finding Slopes of Tangent Lines



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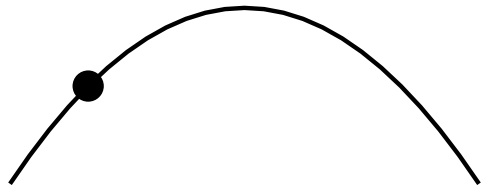
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Slope of tangent line: can't do the same way.

Finding Slopes of Tangent Lines



We NEED limits to find slopes of tangent lines.



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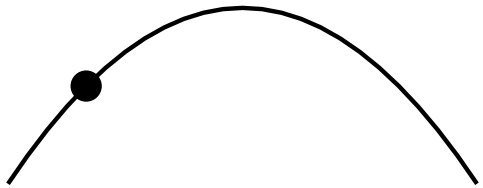
Slope of tangent line: can't do the same way.

If the position of an object at time t is given by $s(t)$, then its instantaneous velocity is given by

Finding Slopes of Tangent Lines



We NEED limits to find slopes of tangent lines.



Slope of secant line: $\frac{\Delta y}{\Delta x}$, $\Delta x \neq 0$.

Slope of tangent line: can't do the same way.

If the position of an object at time t is given by $s(t)$, then its instantaneous velocity is given by

$$\lim_{h \to 0} \frac{s(t+h) - s(t)}{h}$$

Evaluating Limits

$$\text{Let } f(x) = \frac{x^3 + x^2 - x - 1}{x - 1}.$$

We want to evaluate $\lim_{x \to 1} f(x)$.

Evaluating Limits

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What is $f(1)$?

Evaluating Limits

$$\text{Let } f(x) = \frac{x^3 + x^2 - x - 1}{x - 1}.$$

We want to evaluate $\lim_{x \to 1} f(x)$.

What is $f(1)$? DNE (can't divide by zero)

Evaluating Limits

$$\text{Let } f(x) = \frac{x^3 + x^2 - x - 1}{x - 1}.$$

We want to evaluate $\lim_{x \to 1} f(x)$.

Use the tables below to guess $\lim_{x \to 1} f(x)$

x	$f(x)$
0.9	3.61
0.99	3.9601
0.999	3.99600
0.9999	3.99960

x	$f(x)$
1.1	4.41
1.01	4.0401
1.001	4.00400
1.0001	4.00040

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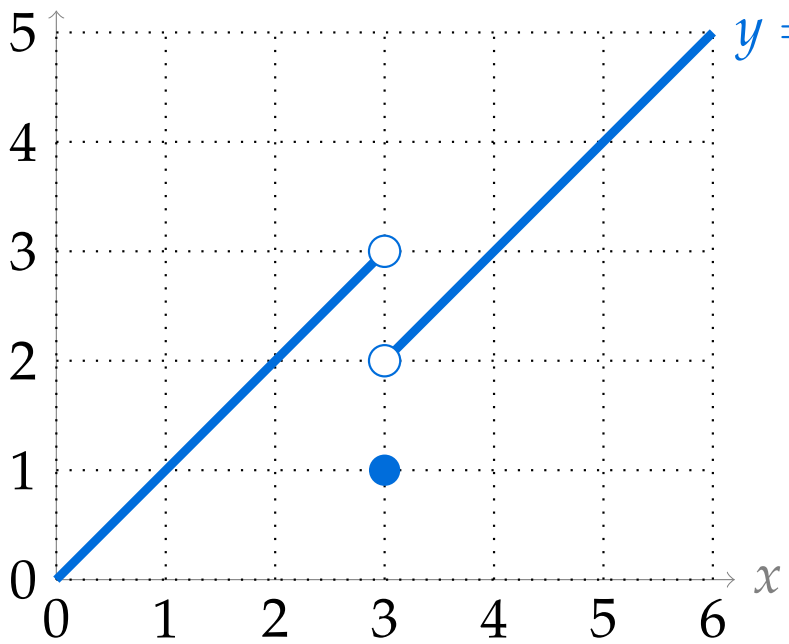
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$$\boxed{\lim_{x \to 1} f(x) = 4}$$

One-Sided Limits

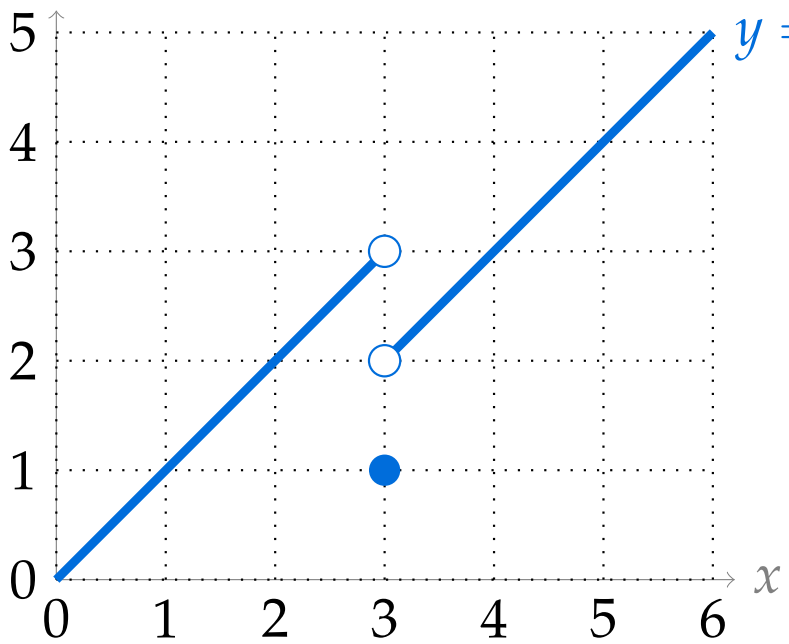


$$f(x) = \begin{cases} x & \text{if } x < 3 \\ 1 & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \end{cases}$$

What do you think $\lim_{x \to 3} f(x)$ should be?

Q

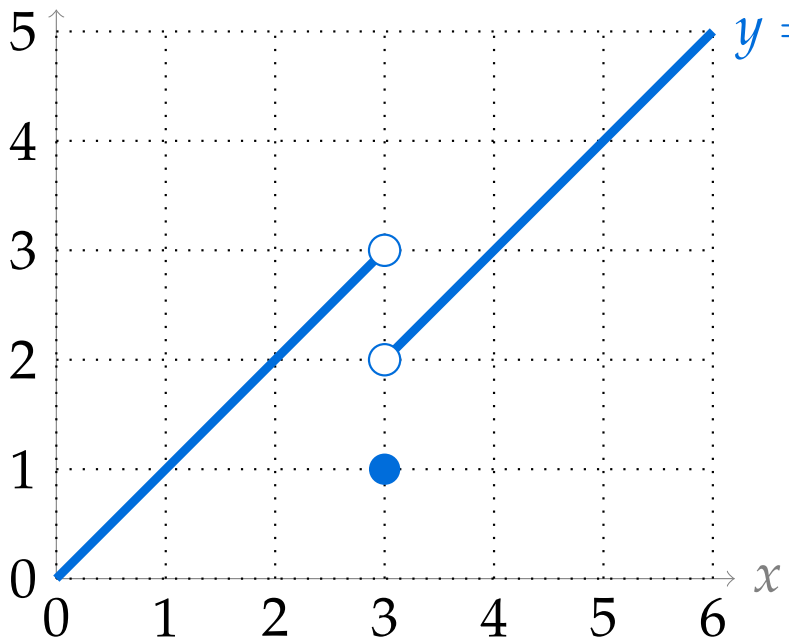
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One-Sided Limits

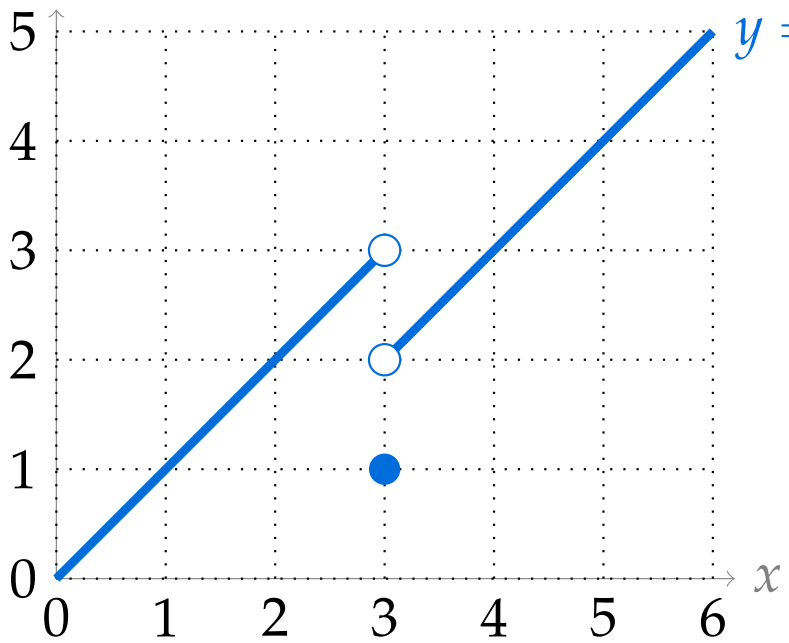


$$f(x) = \begin{cases} x & \text{if } x < 3 \\ 1 & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \end{cases}$$

Evaluate: $\underbrace{\lim_{x \to 3^-} f(x)}_{\text{from the left}}$ $\underbrace{\lim_{x \to 3^+} f(x)}_{\text{from the right}}$

Q

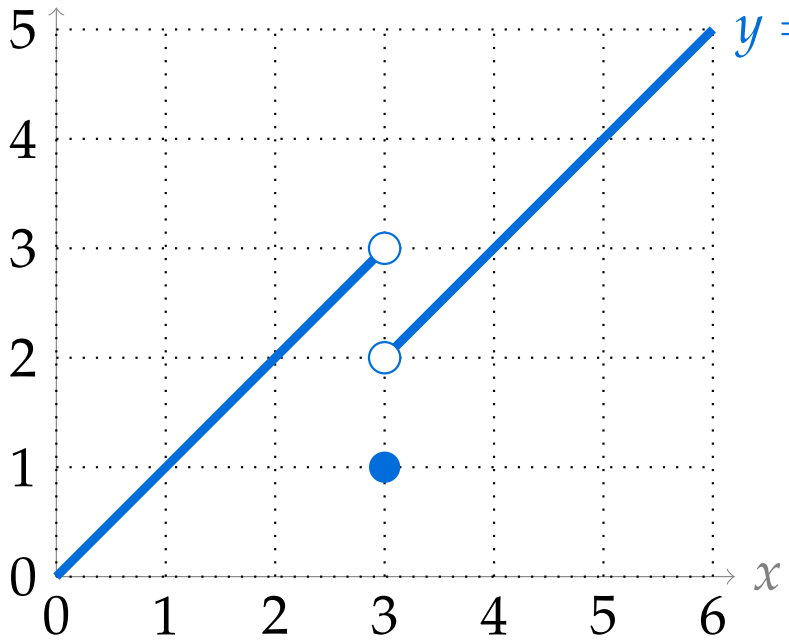
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Evaluate: $\underbrace{\lim_{x \to 3^-} f(x)}_{\text{from the left}} = 3$ $\underbrace{\lim_{x \to 3^+} f(x)}_{\text{from the right}}$

One-Sided Limits



$$f(x) = \begin{cases} x & \text{if } x < 3 \\ 1 & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \end{cases}$$

Evaluate: $\underbrace{\lim_{x \to 3^-} f(x)}_{\text{from the left}} = 3$ $\underbrace{\lim_{x \to 3^+} f(x)}_{\text{from the right}} = 2$

Definition 1.3.7

The limit as x goes to a **from the left** of $f(x)$ is written

$$\lim_{x \to a^-} f(x)$$

We only consider values of x that are less than a.

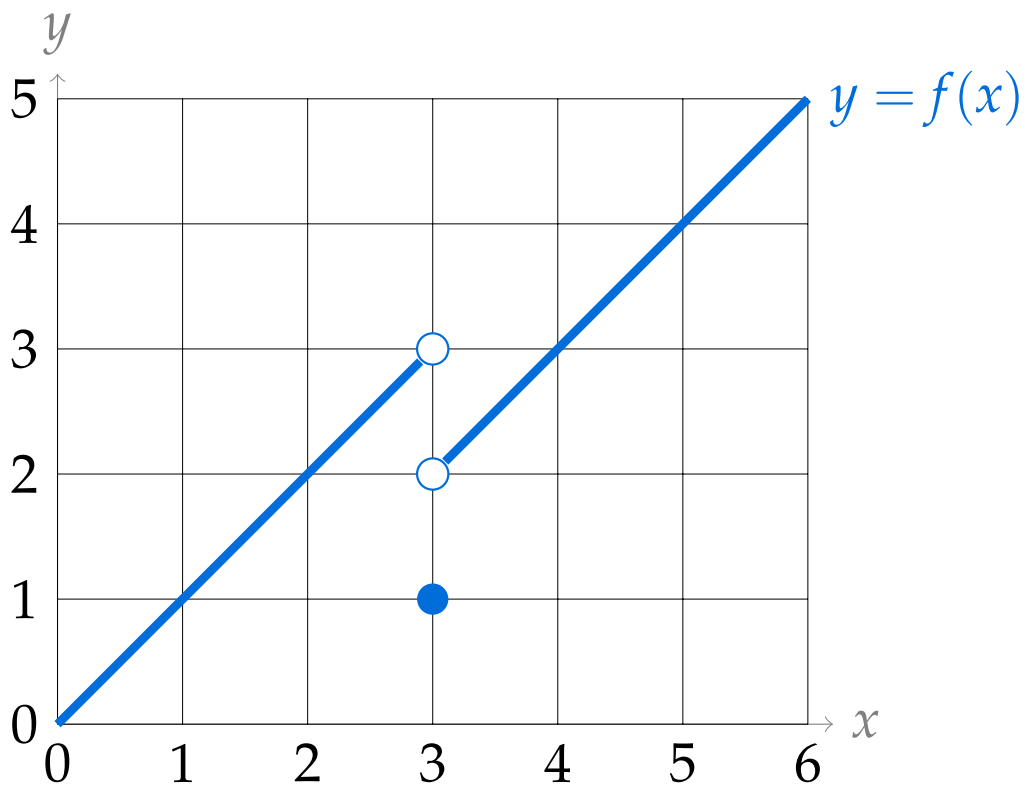
The limit as x goes to a **from the right** of $f(x)$ is written

$$\lim_{x \to a^+} f(x)$$

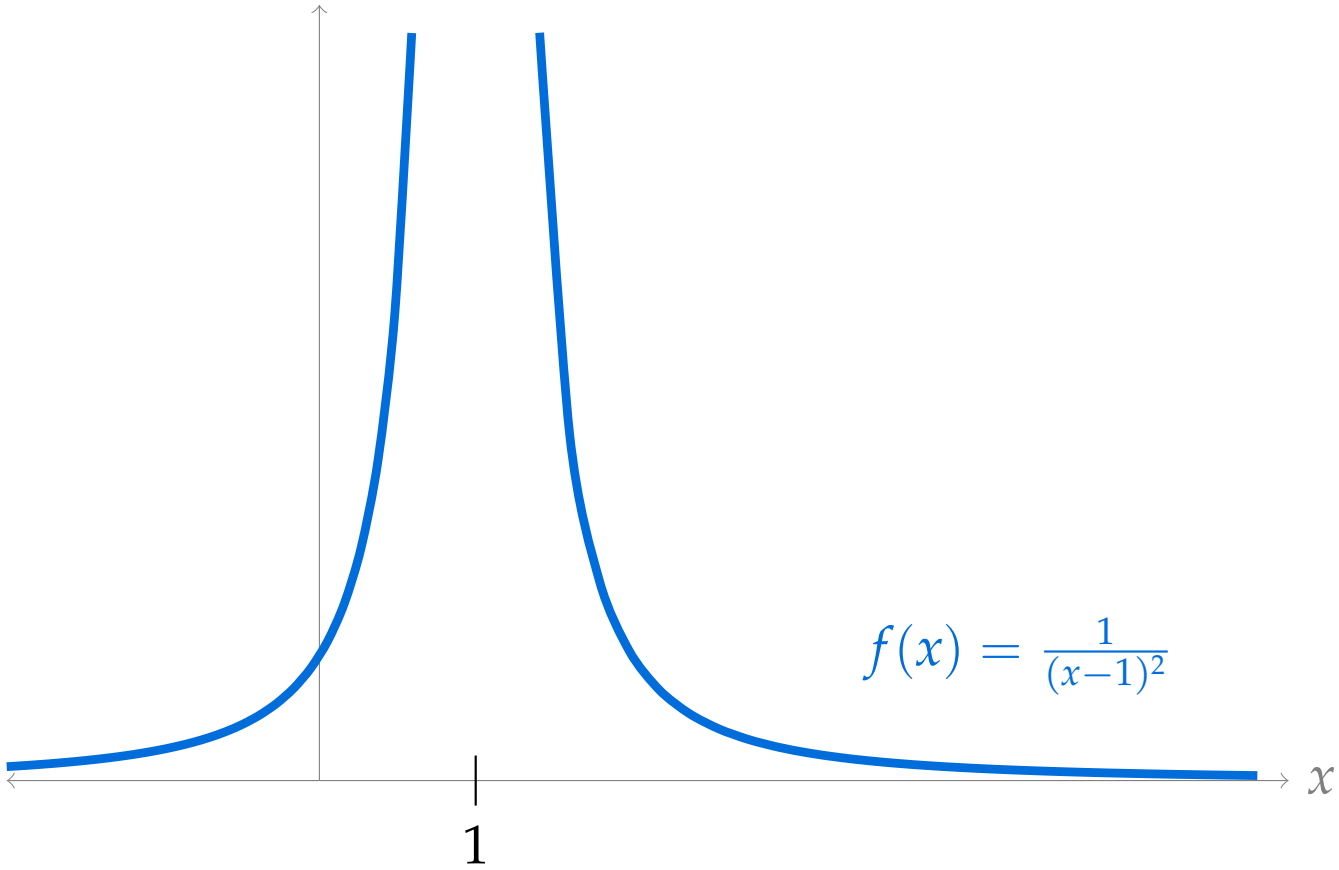
We only consider values of x greater than a.

Theorem 1.3.8

In order for $\lim_{x \to a} f(x)$ to exist, both one-sided limits must exist and be equal.

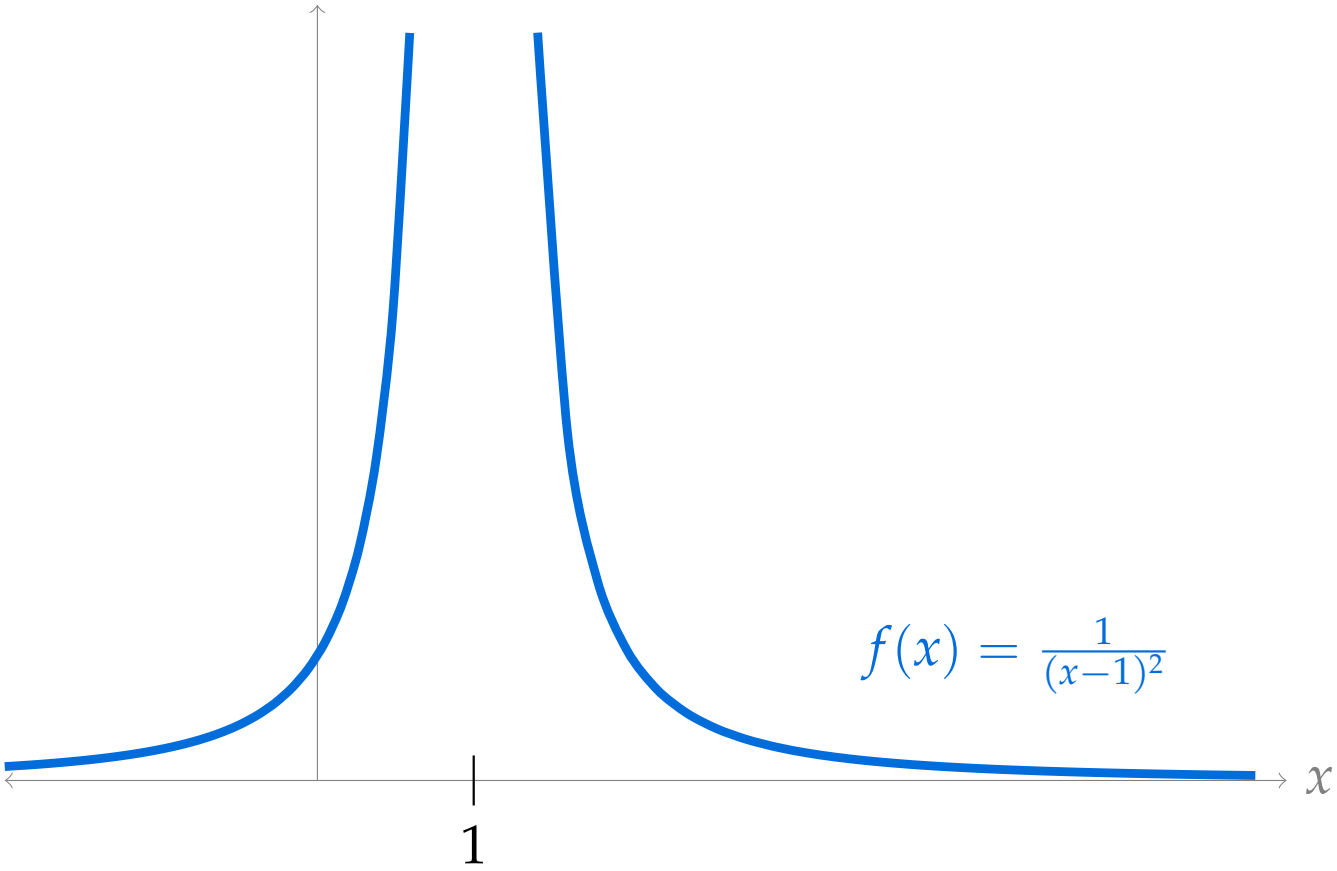


Consider the function $f(x) = \dfrac{1}{(x-1)^2}$. For what value(s) of x is $f(x)$ not defined?



Based on the graph below, what would you like to write for:

$$\lim_{x \to 1} f(x) =$$

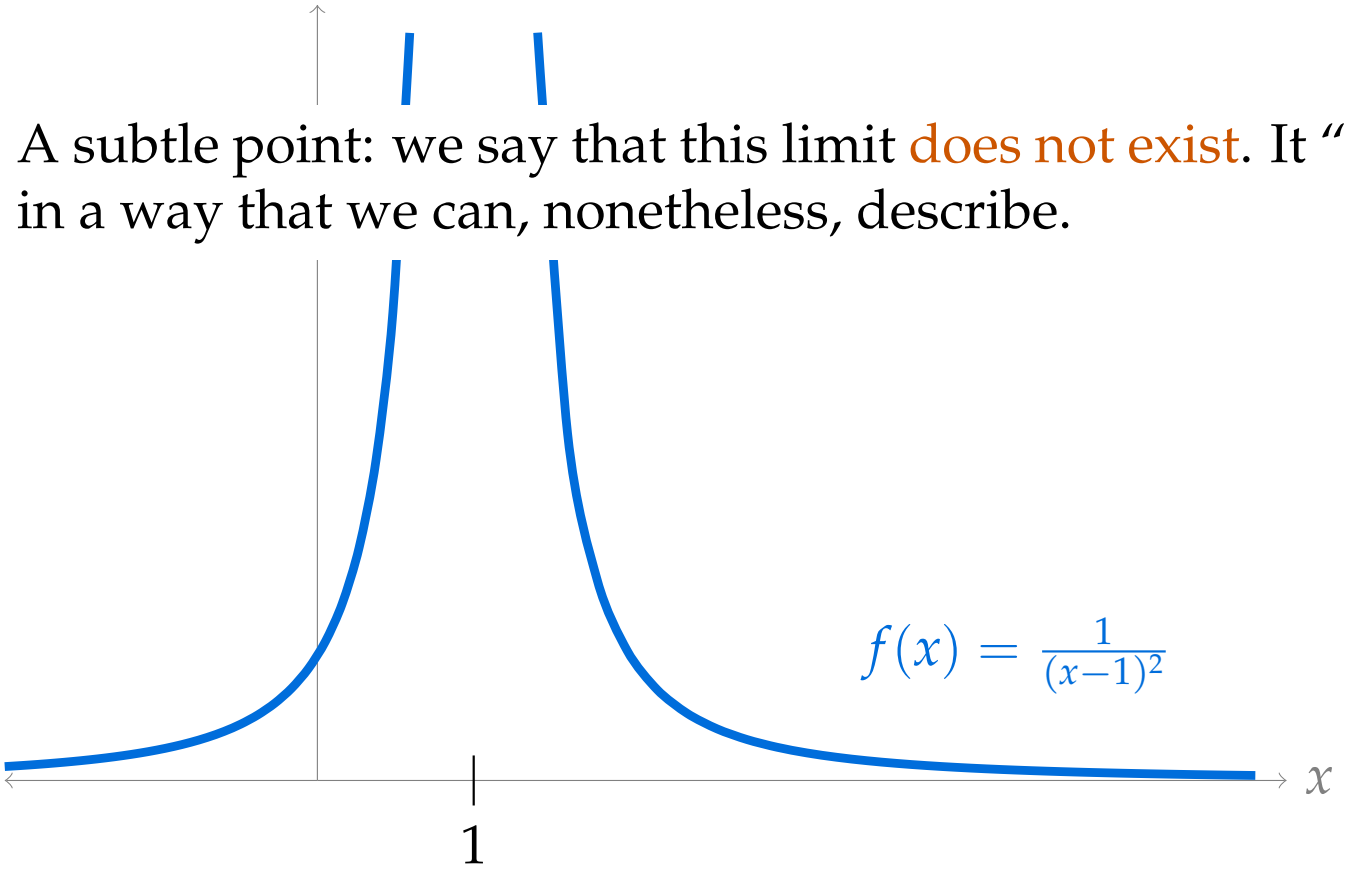


ans

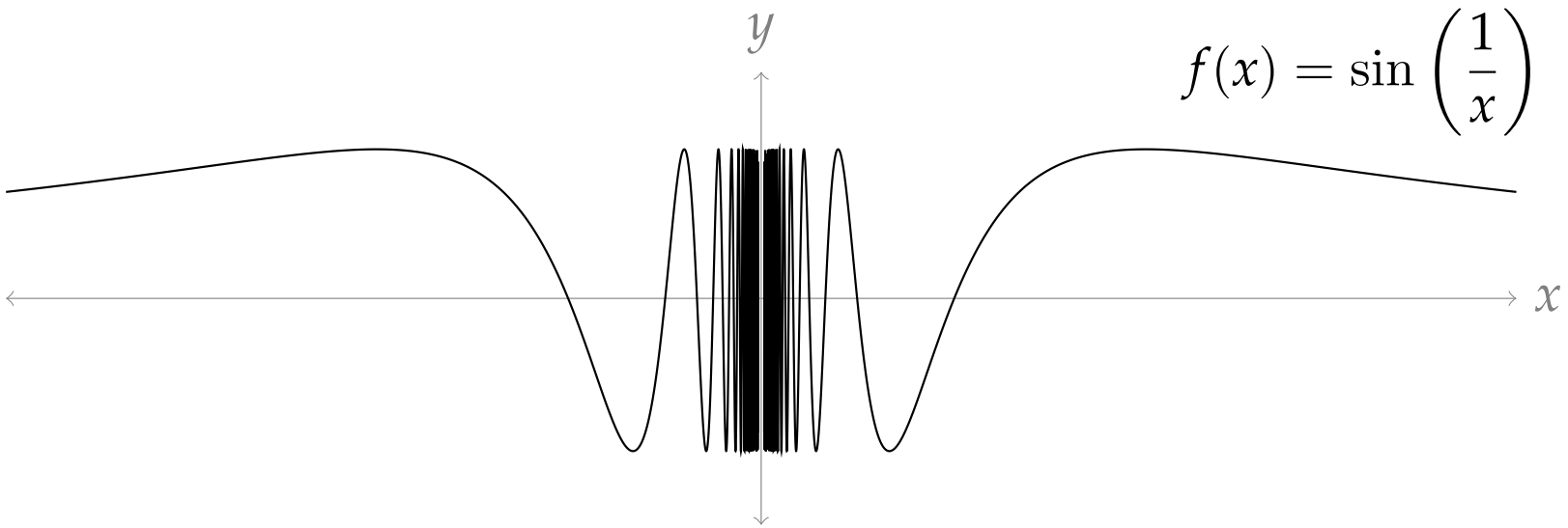
Based on the graph below, what would you like to write for:

$$\lim_{x \to 1} f(x) = \infty$$

A subtle point: we say that this limit does not exist. It "does not exist" in a way that we can, nonetheless, describe.

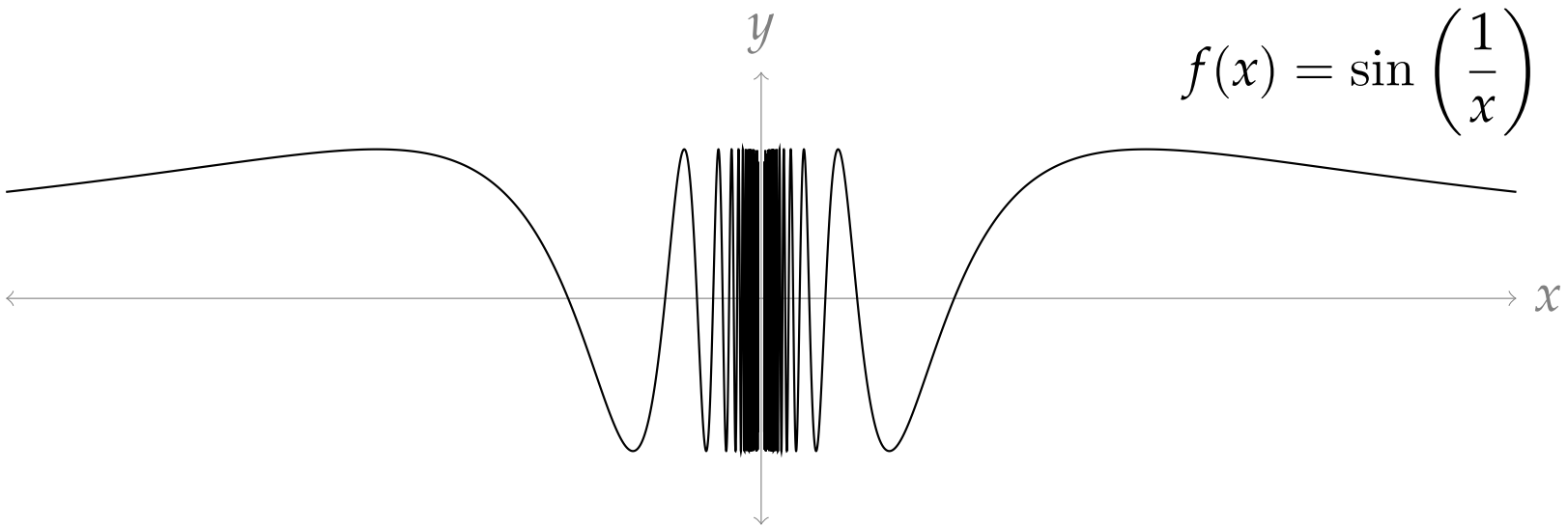


A Stranger Limit Example



What is $\lim_{x \to \infty} f(x)$?

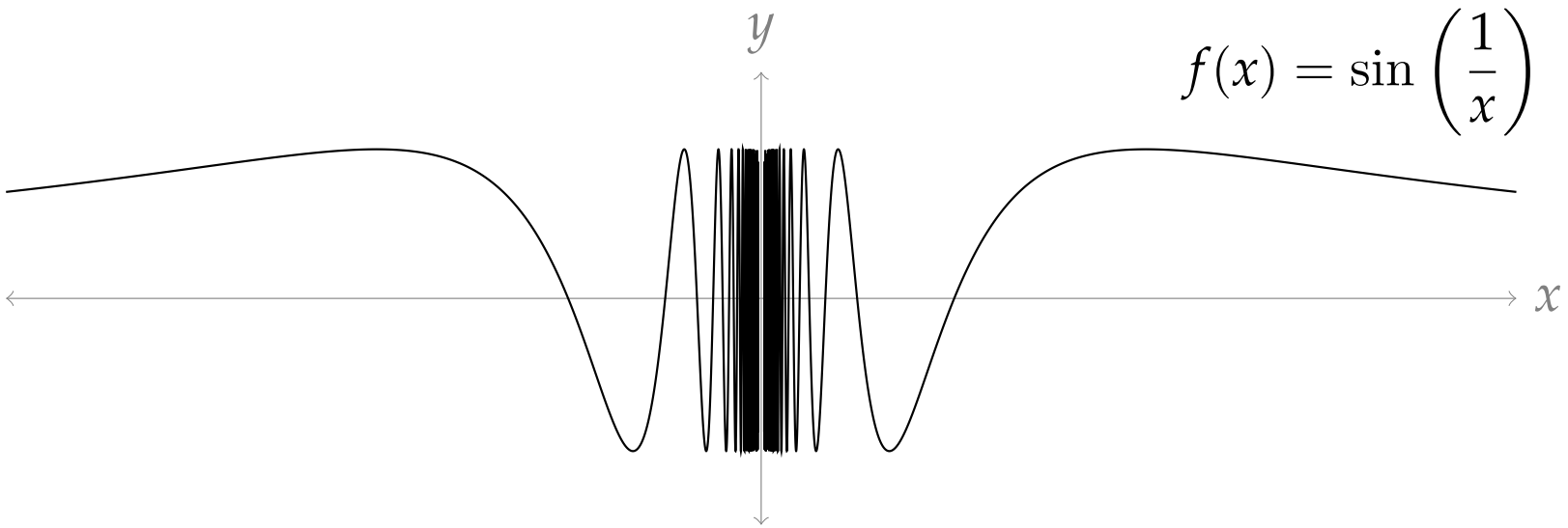
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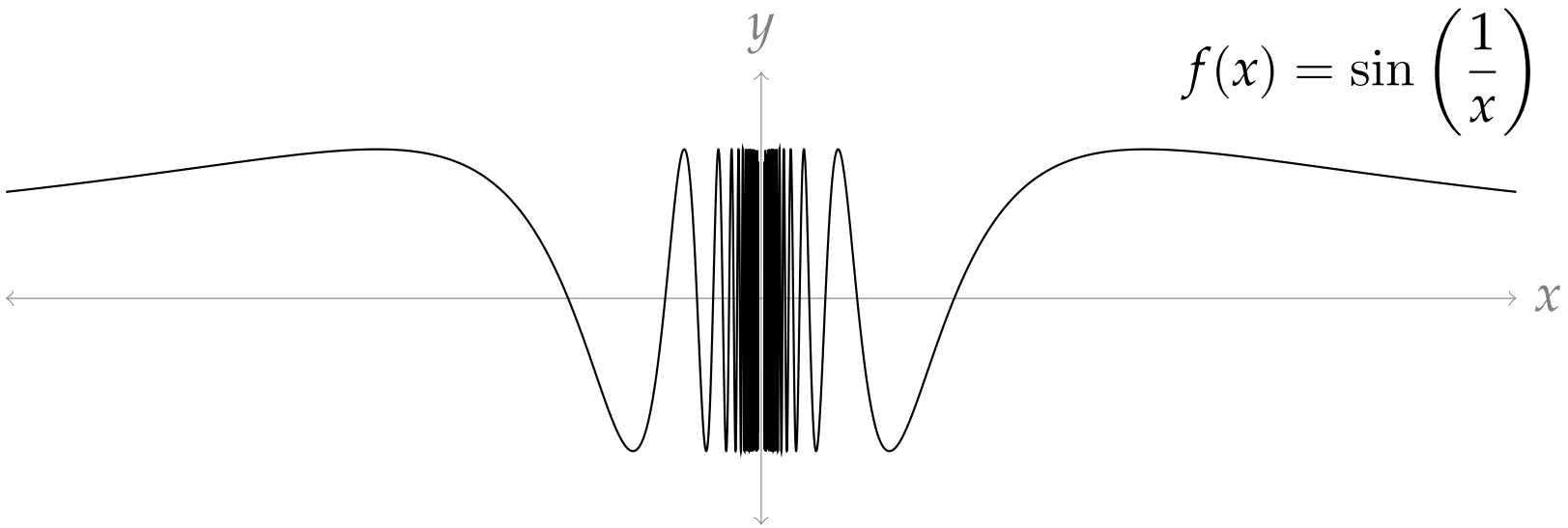
$\lim_{x \to \infty} f(x) = 0$

A Stranger Limit Example



What is $\lim_{x \to 0} f(x)$?

A Stranger Limit Example

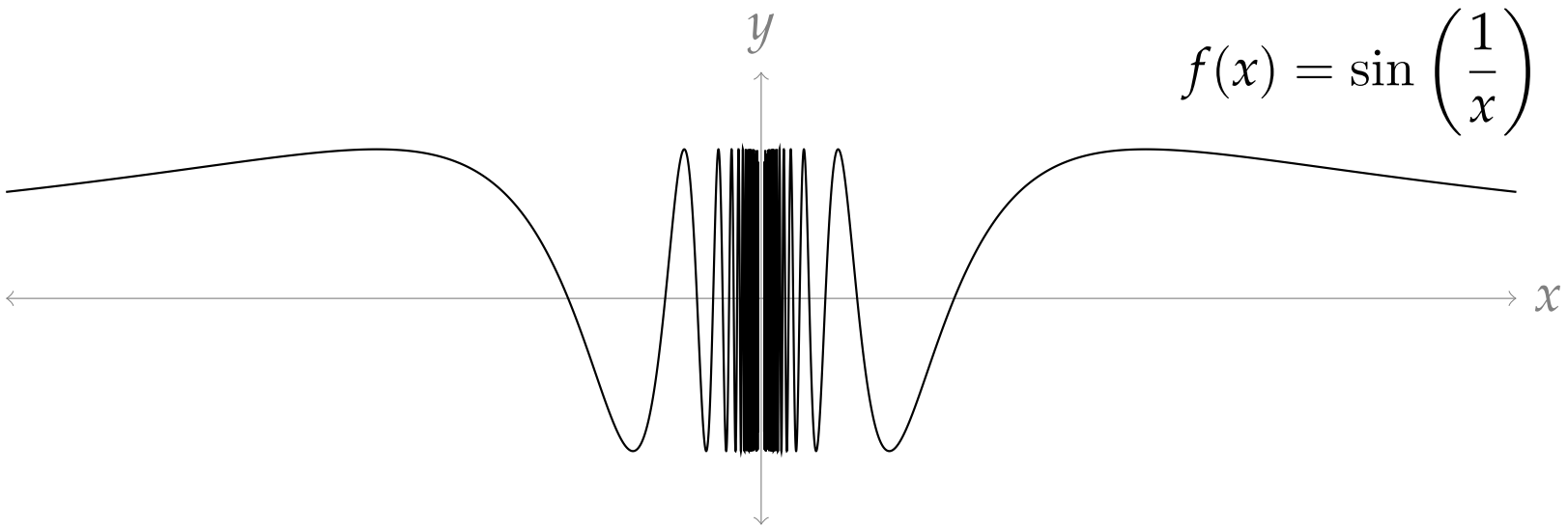


What is $\lim_{x \to 0} f(x)$?

$\lim_{x \to 0} f(x)$ does not exist.

We can call this behaviour "infinite wiggling."

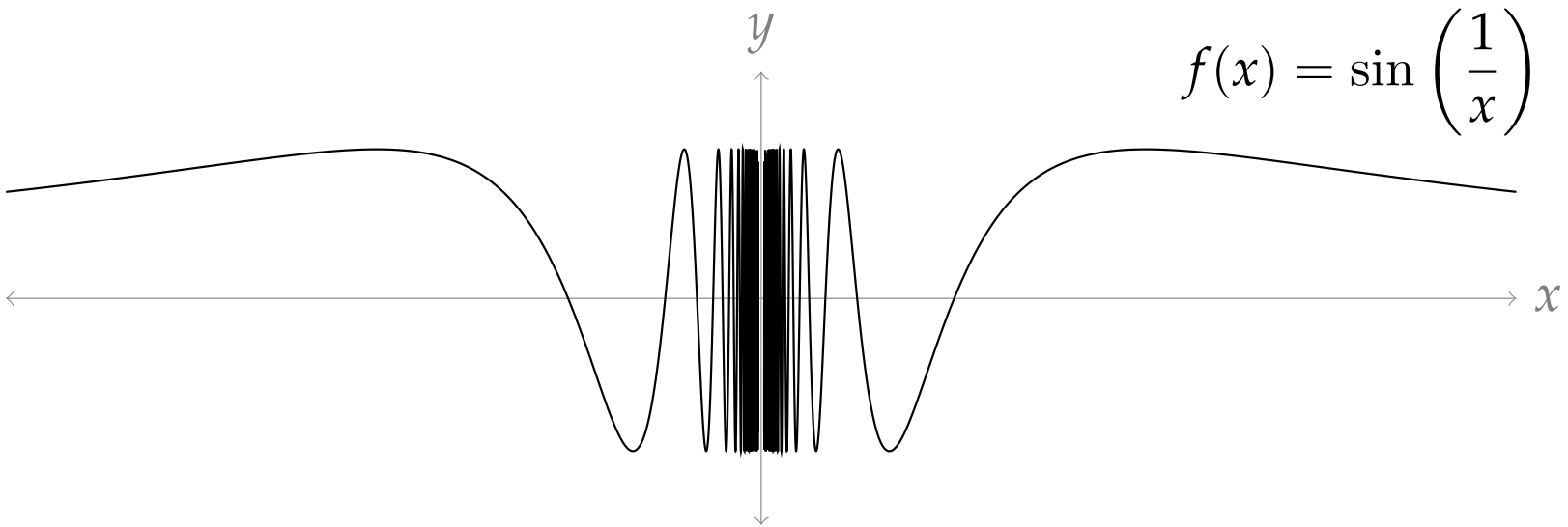
A Stranger Limit Example



What is $\lim_{x \to \pi} f(x)$?

Q

A Stranger Limit Example

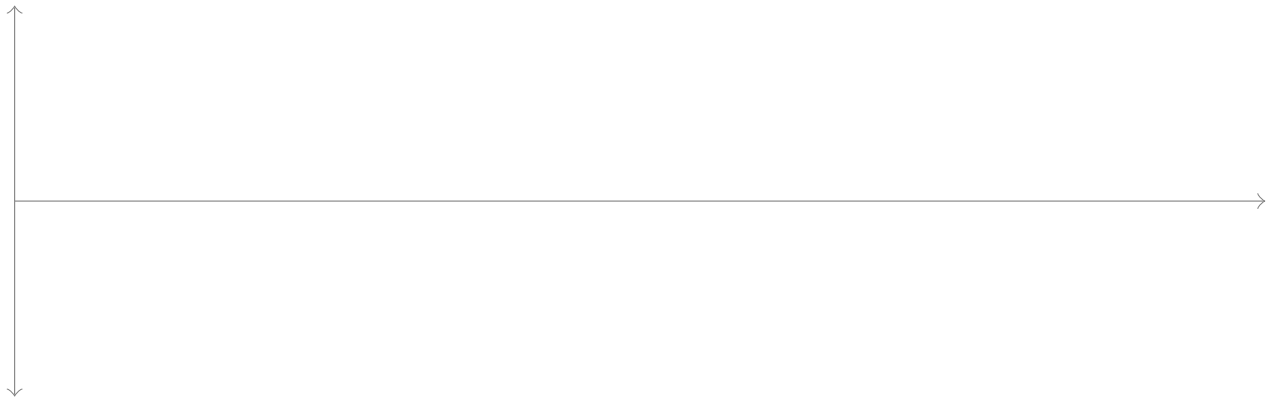


What is $\lim_{x \to \pi} f(x)$?

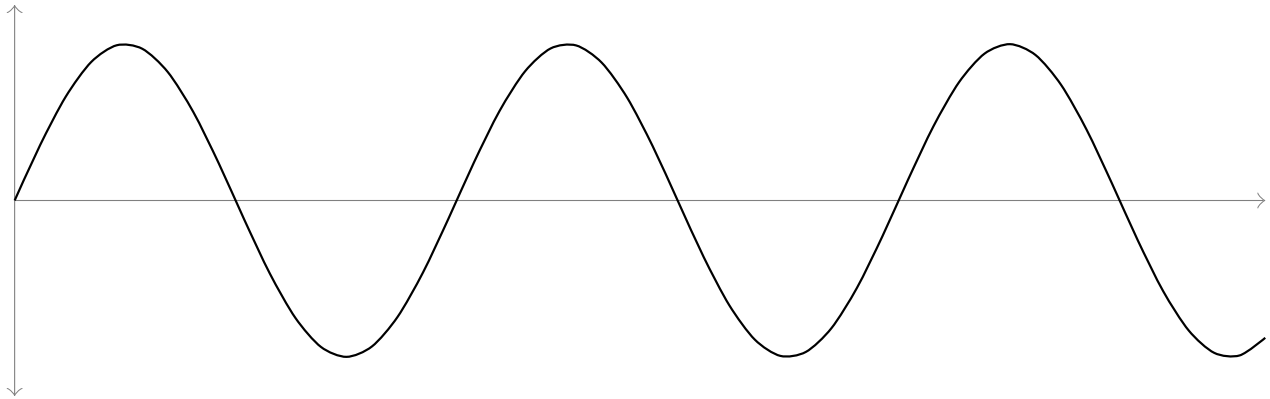
$\lim_{x \to \pi} f(x) = \sin\left(\frac{1}{\pi}\right)$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



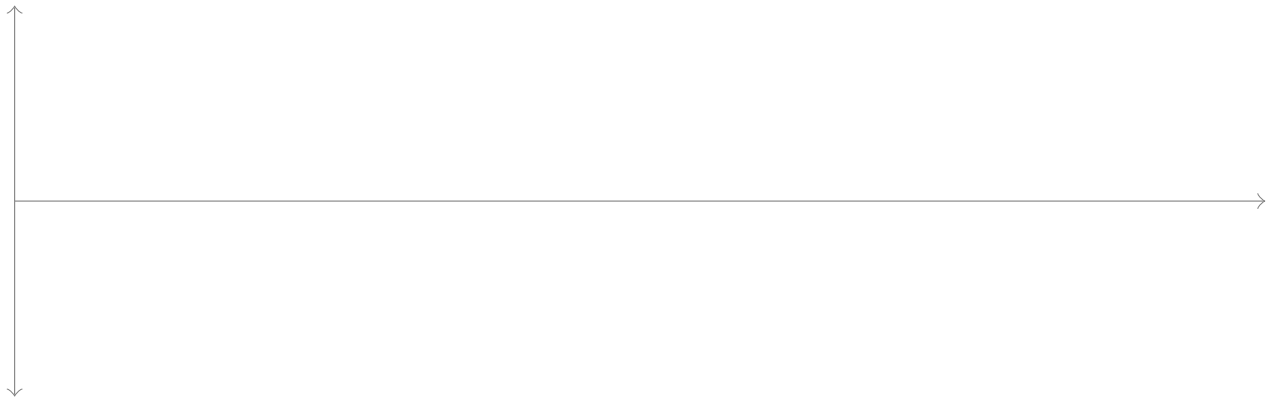
$y = \sin \frac{1}{x}$

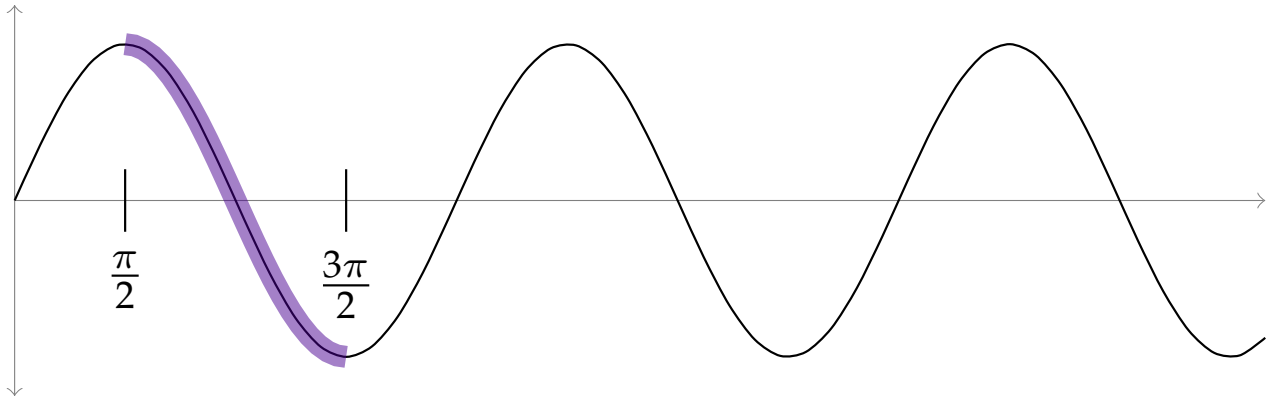


$y = \sin \theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

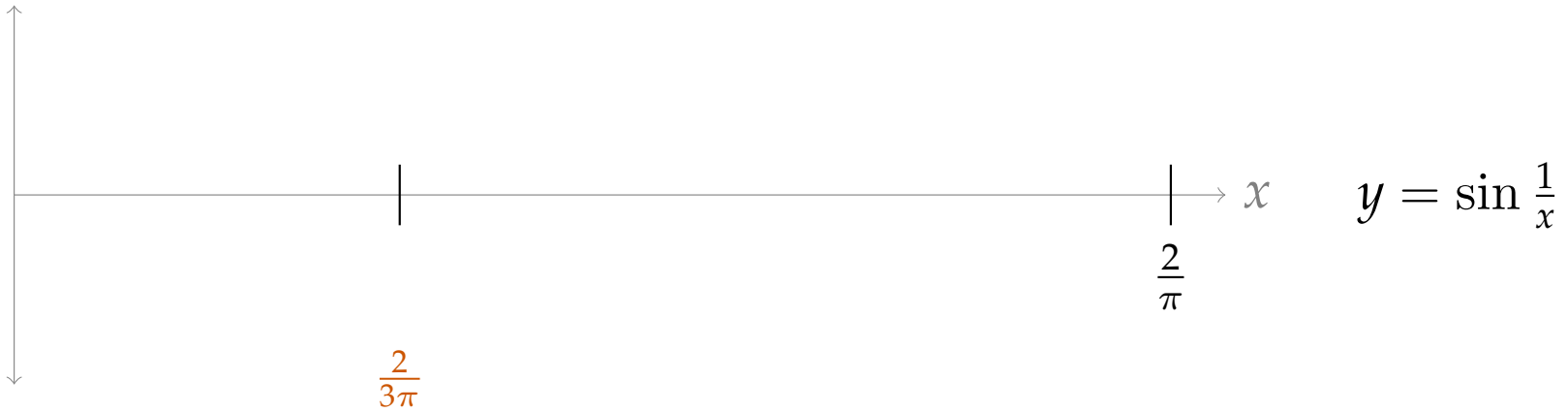
SKIP SKETCHING

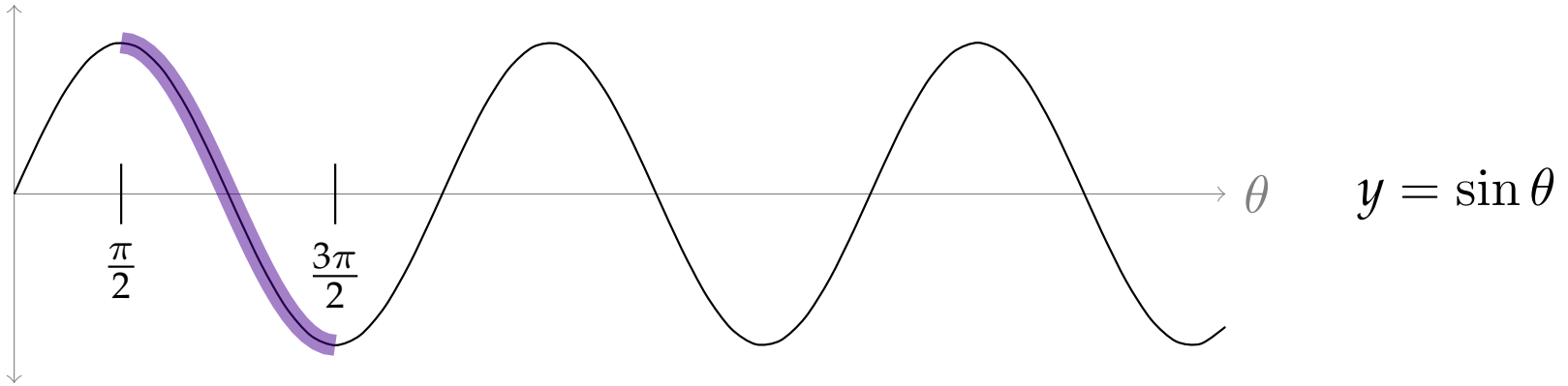




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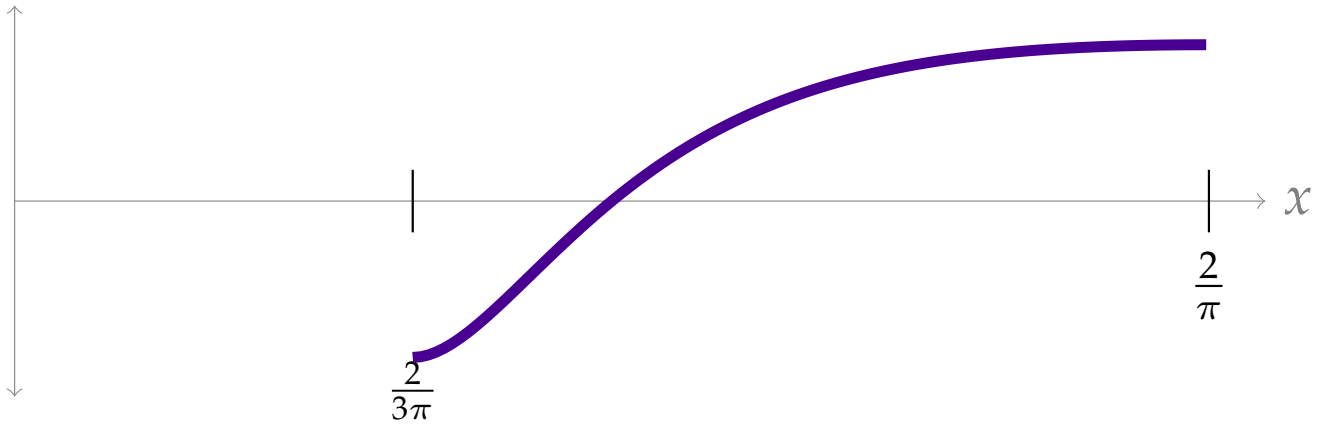
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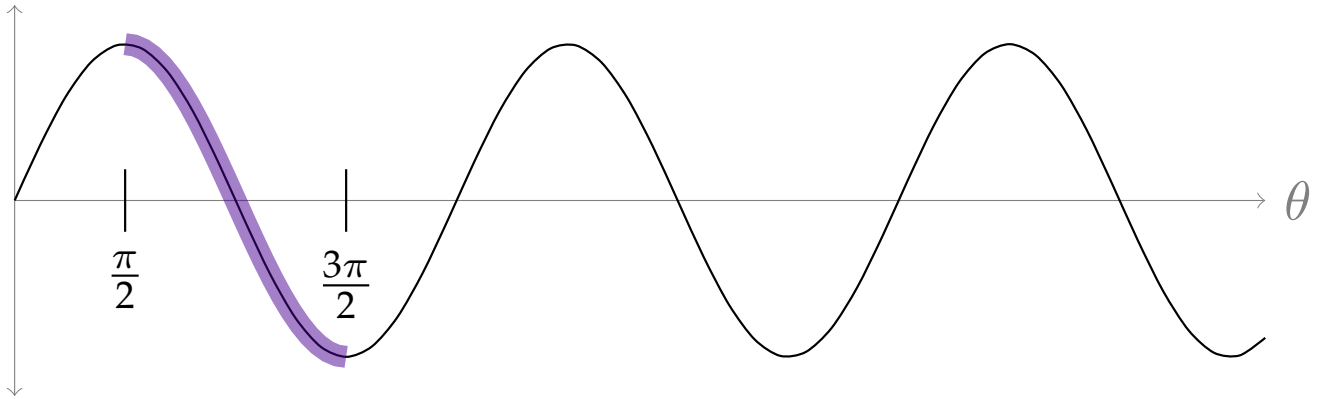


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SKIP SKETCHING



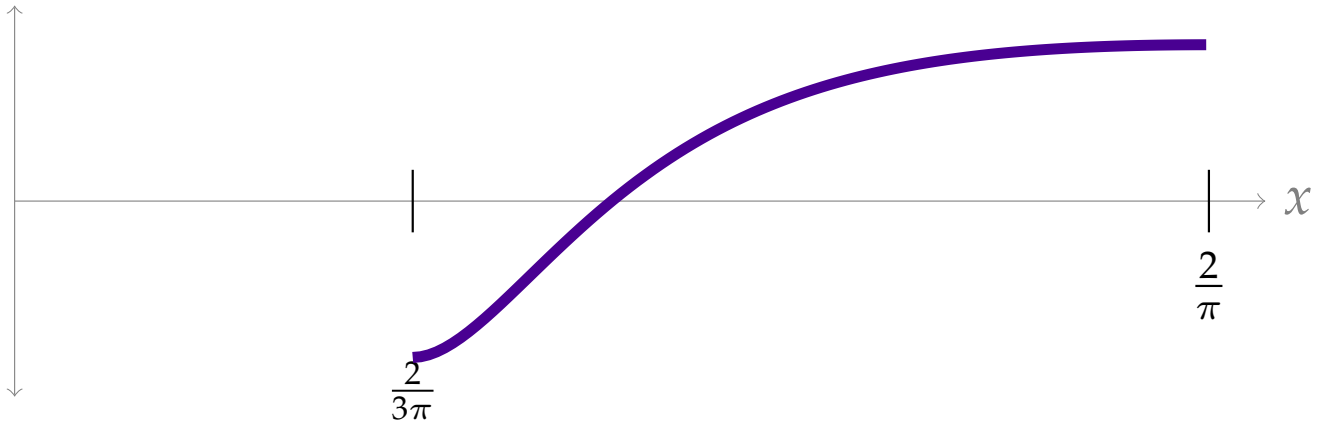
$y = \sin\frac{1}{x}$



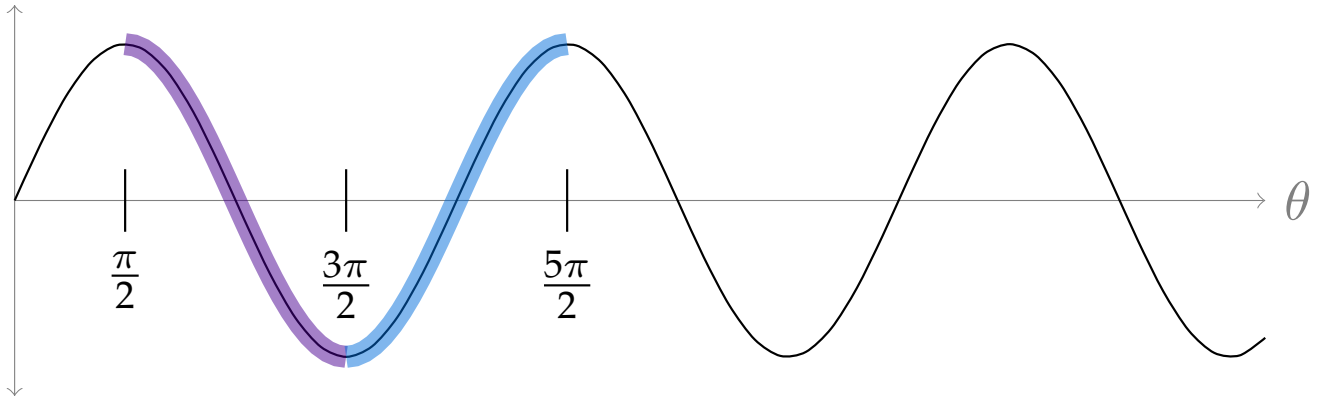
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



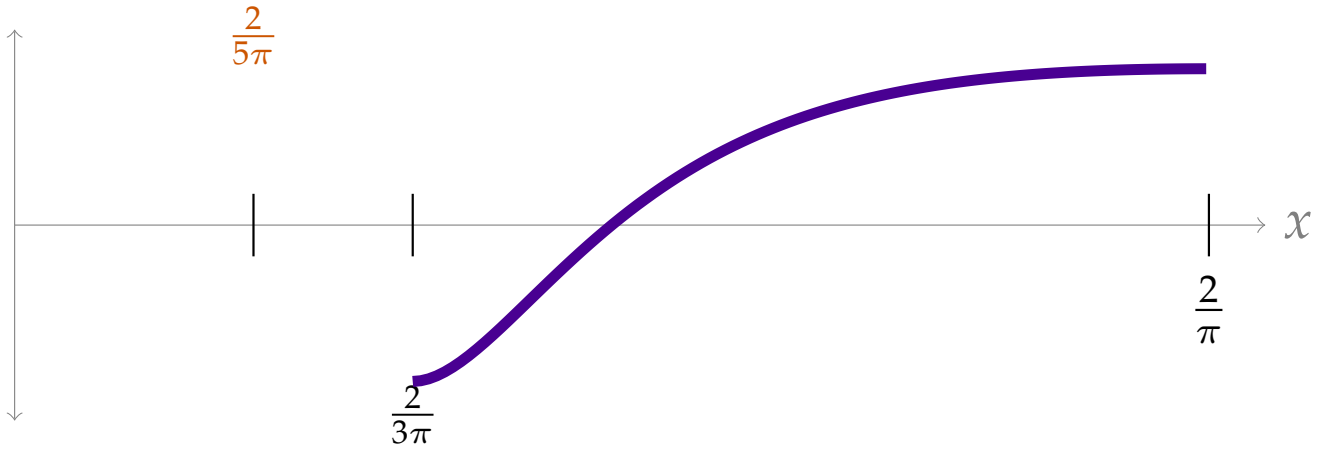
$y = \sin\frac{1}{x}$



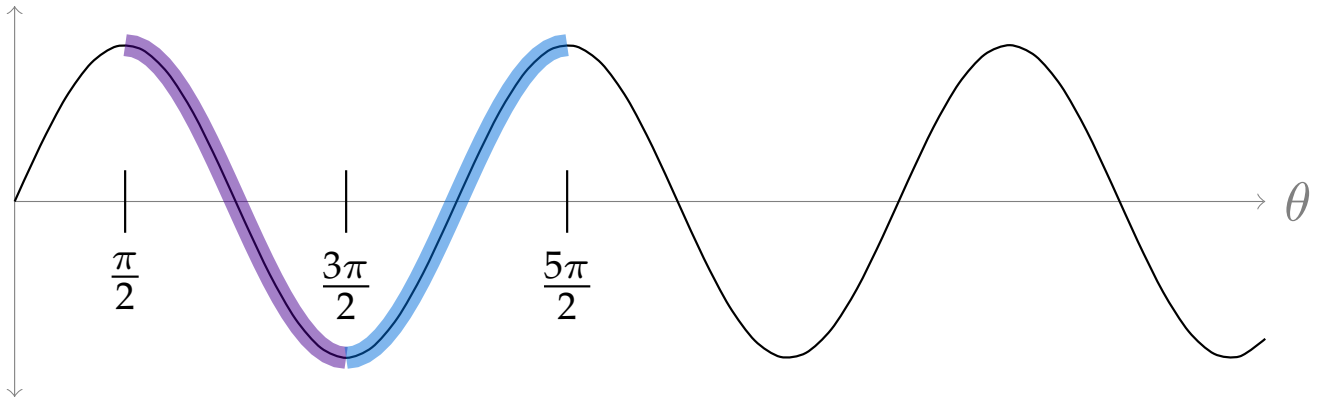
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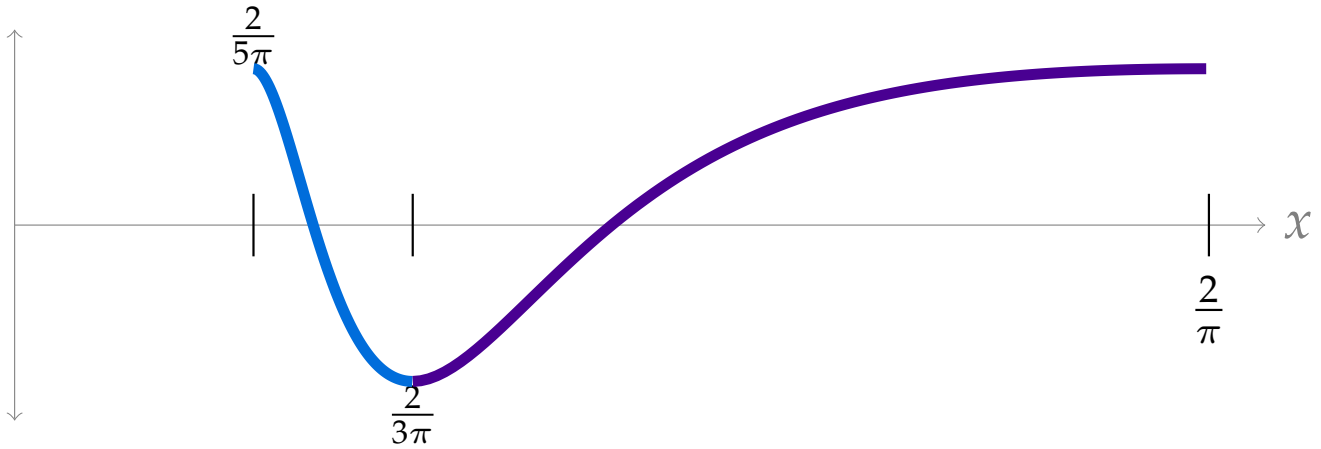
$y = \sin\frac{1}{x}$



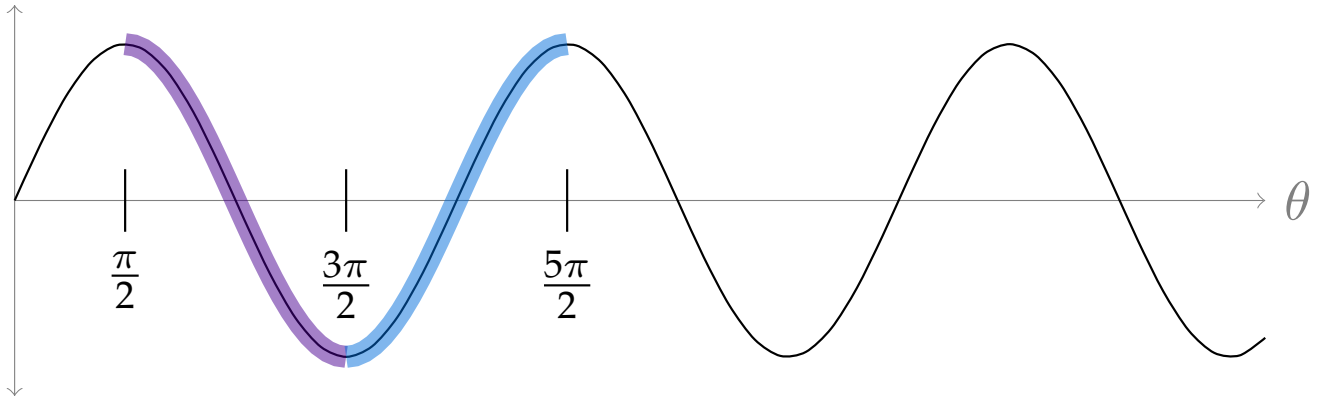
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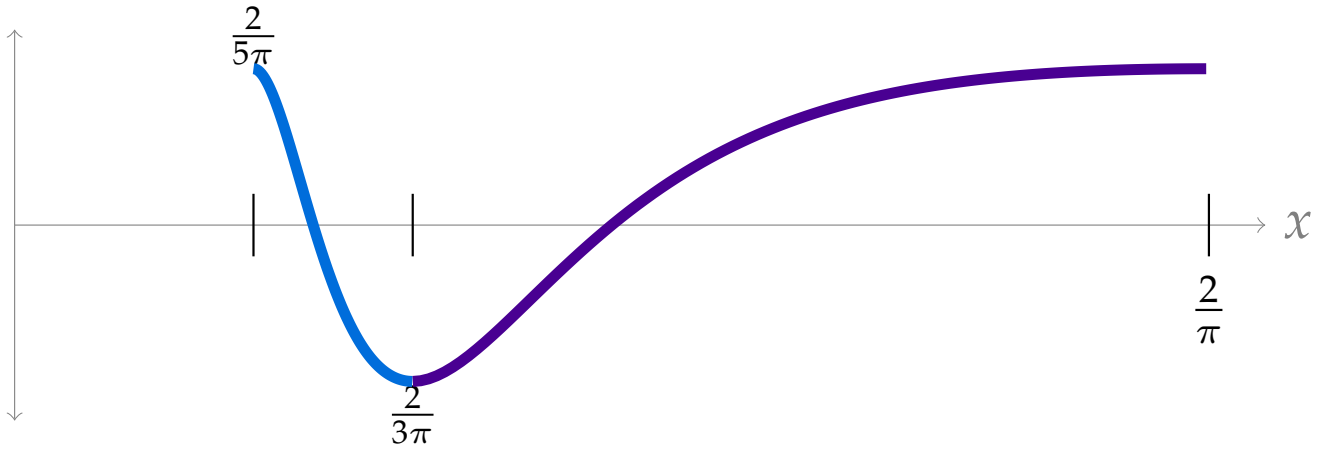
$y = \sin\frac{1}{x}$



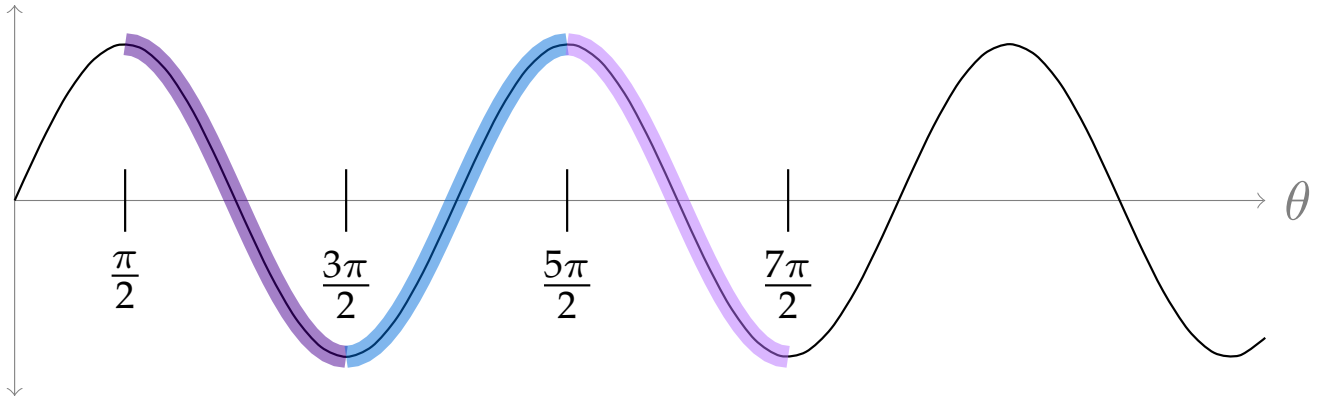
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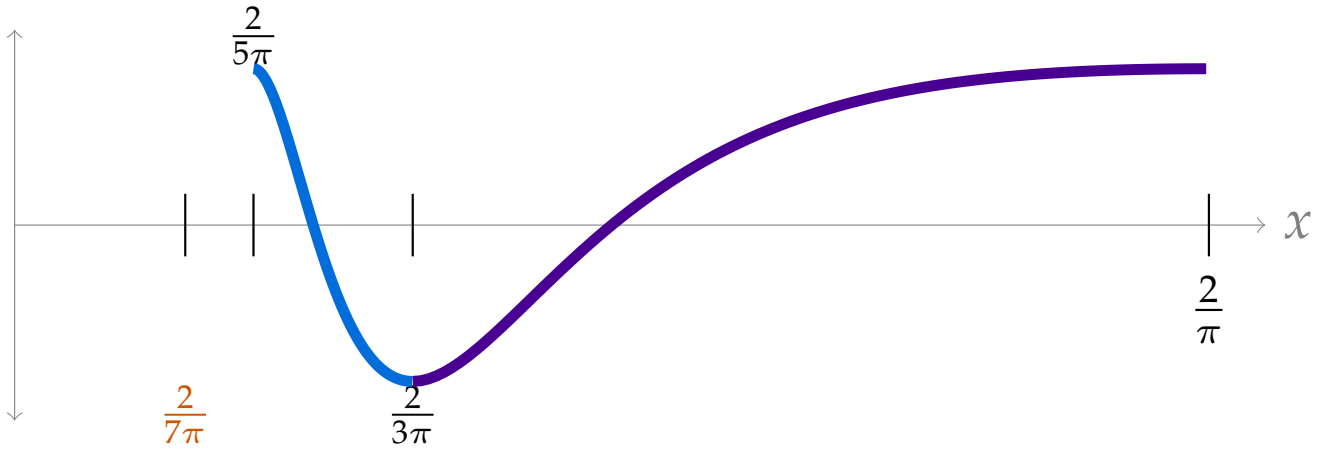
$y = \sin\frac{1}{x}$



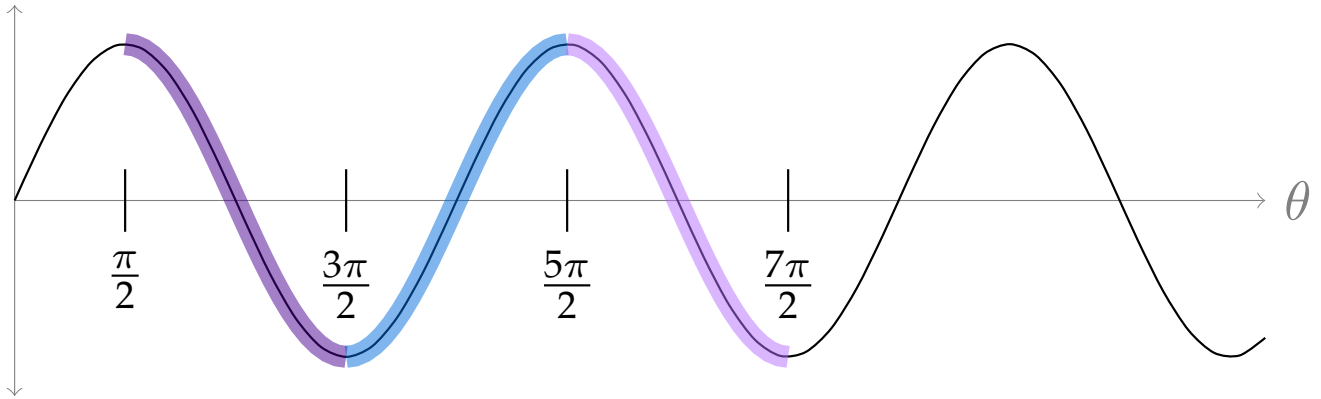
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



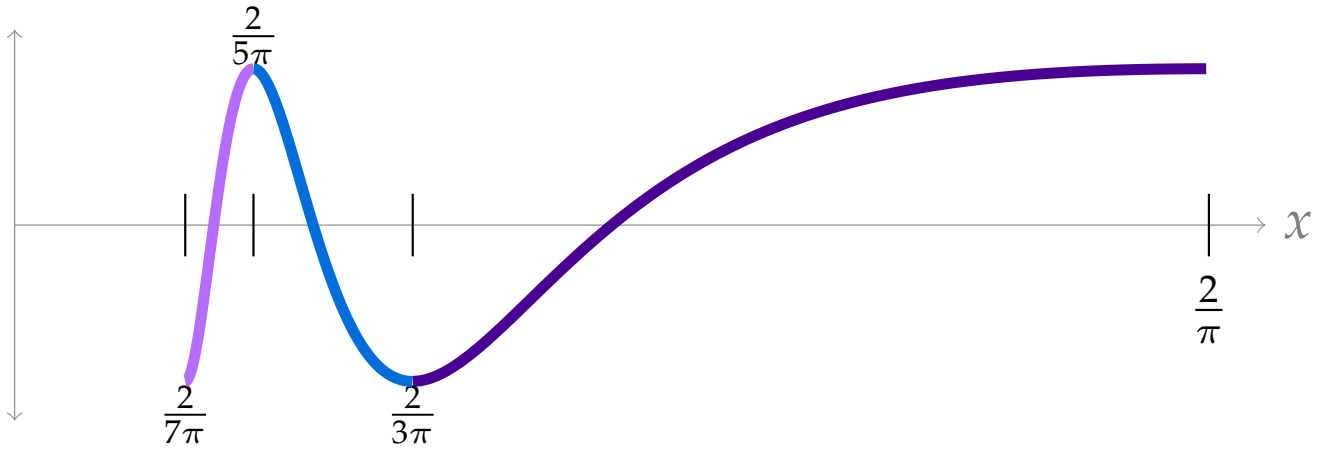
$y = \sin\frac{1}{x}$



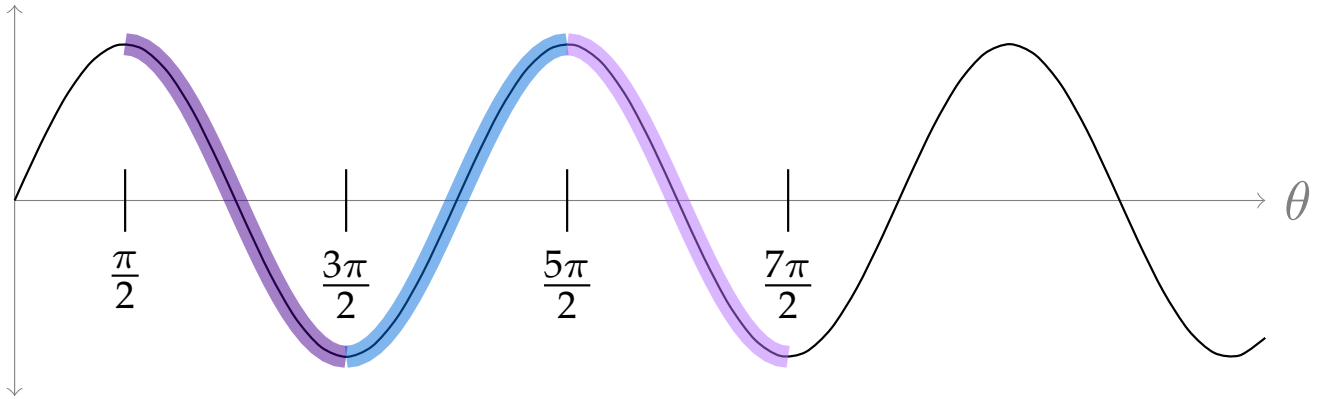
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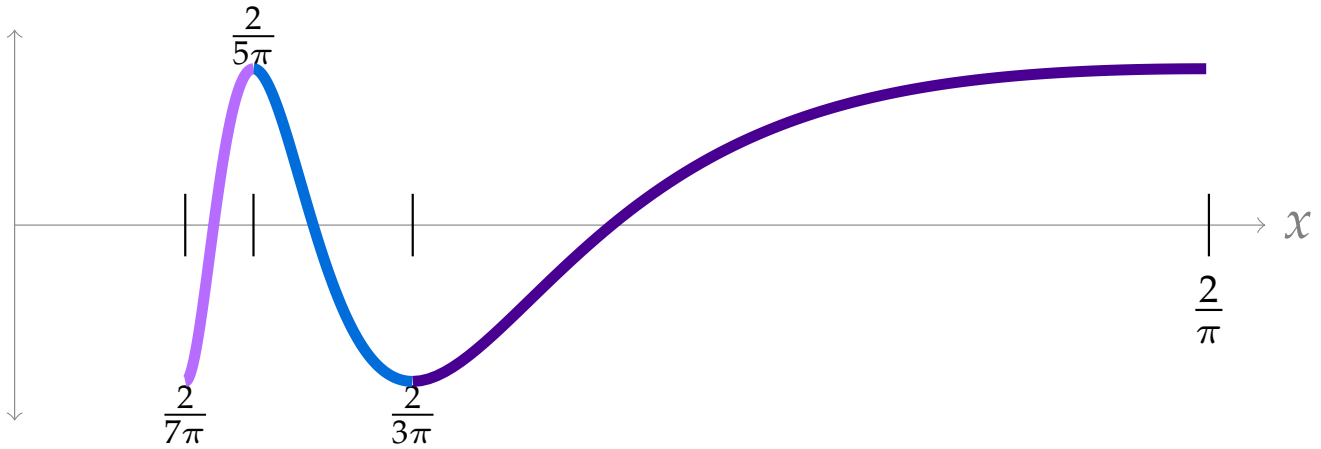
$y = \sin\frac{1}{x}$



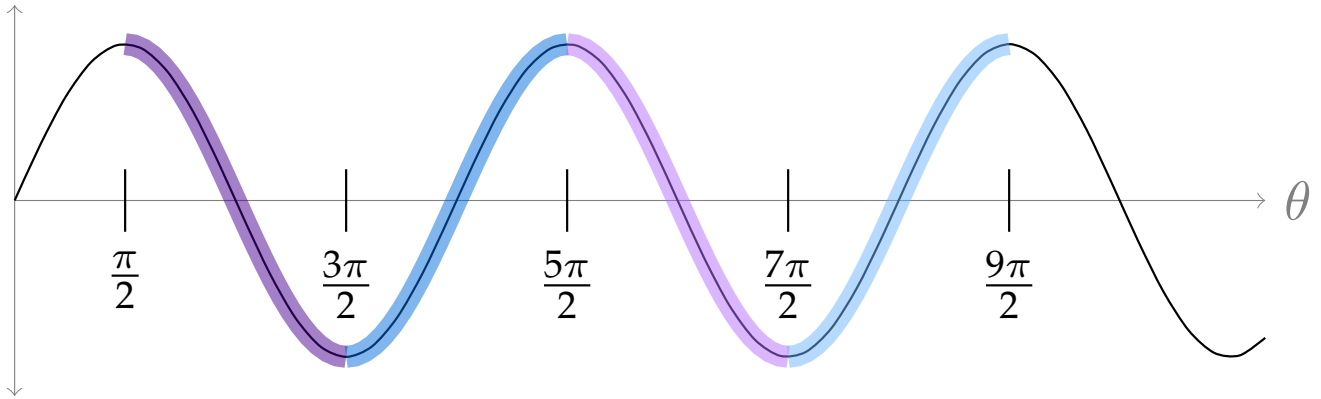
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



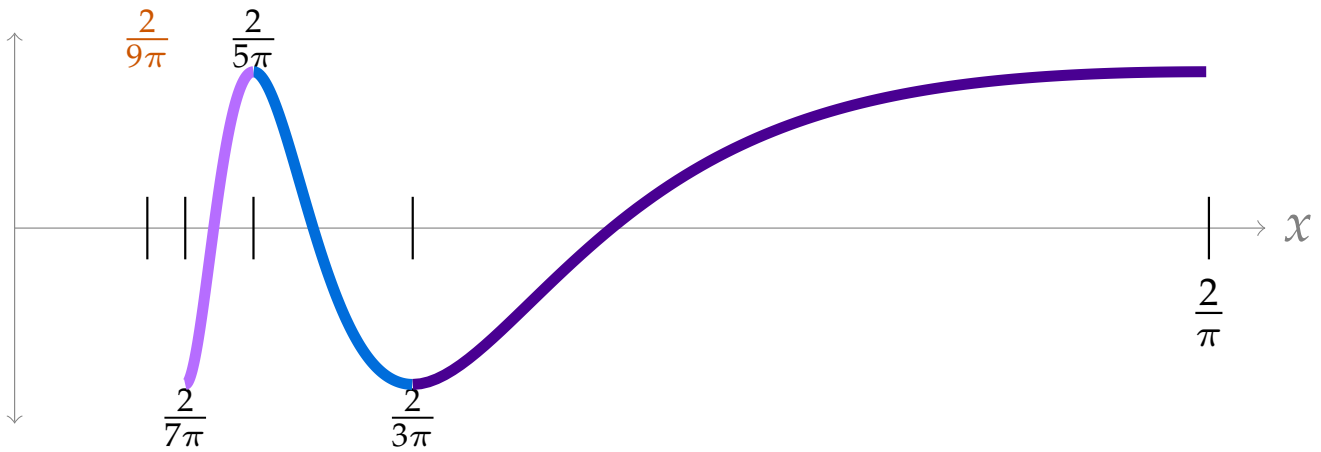
$y = \sin\frac{1}{x}$



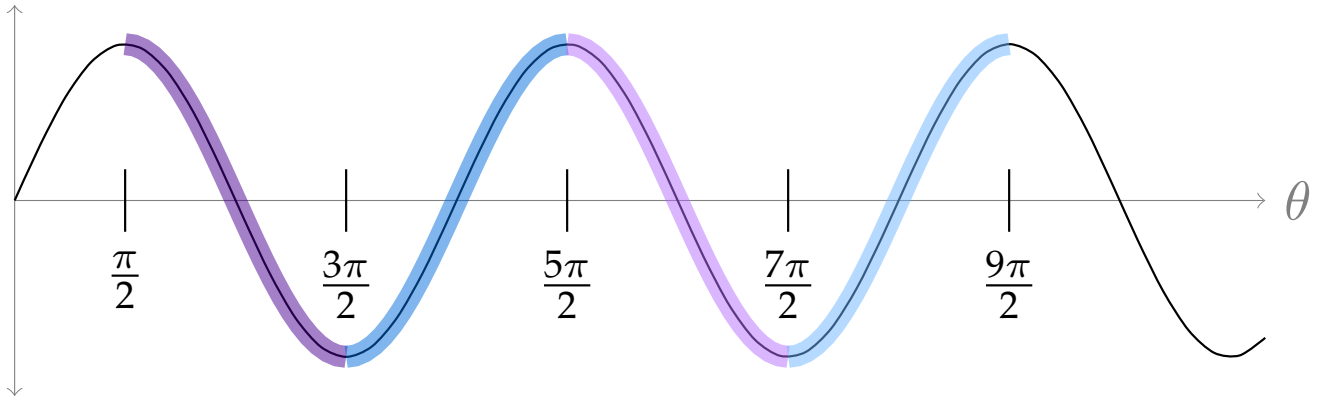
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SKIP SKETCHING



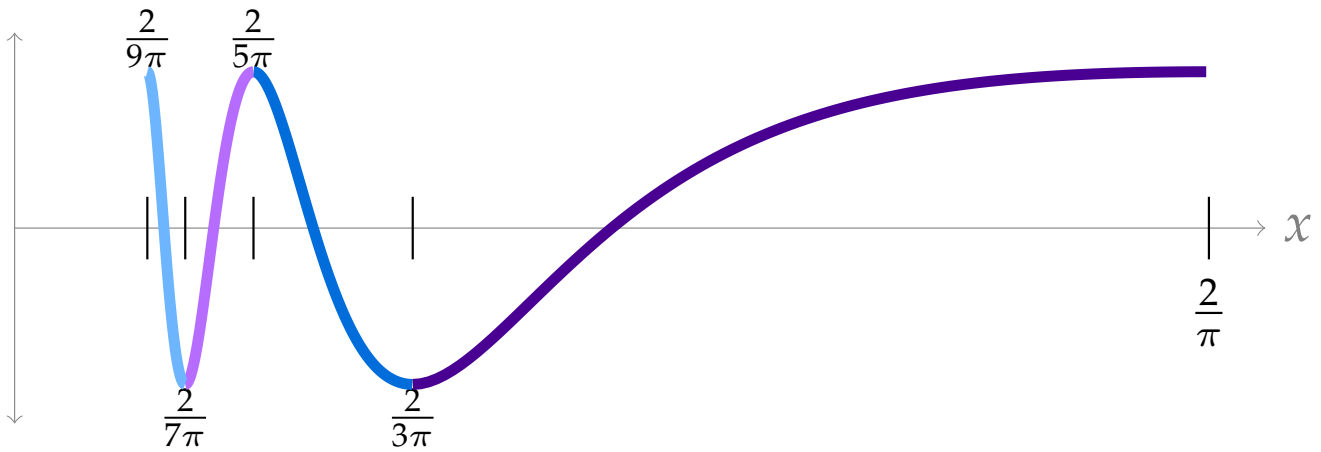
$y = \sin\frac{1}{x}$



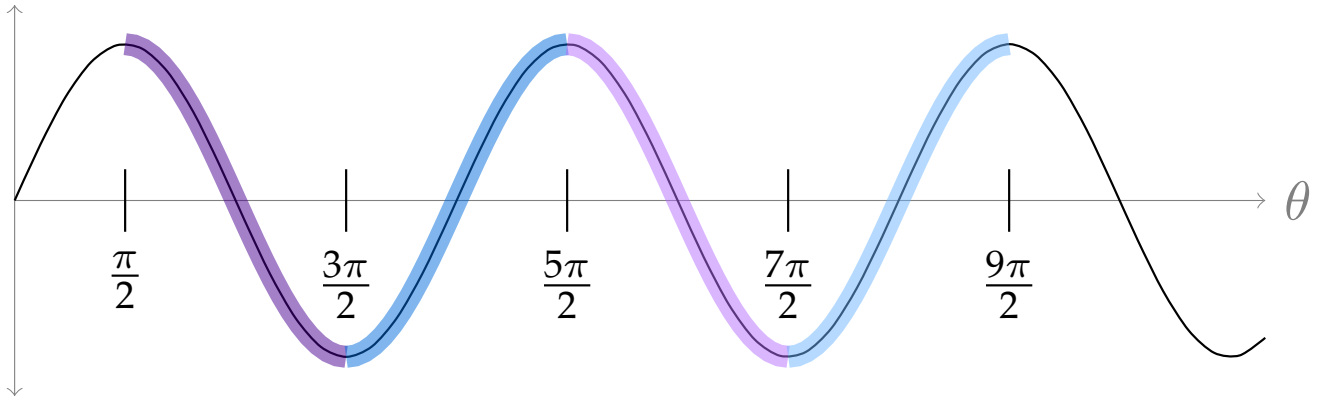
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Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



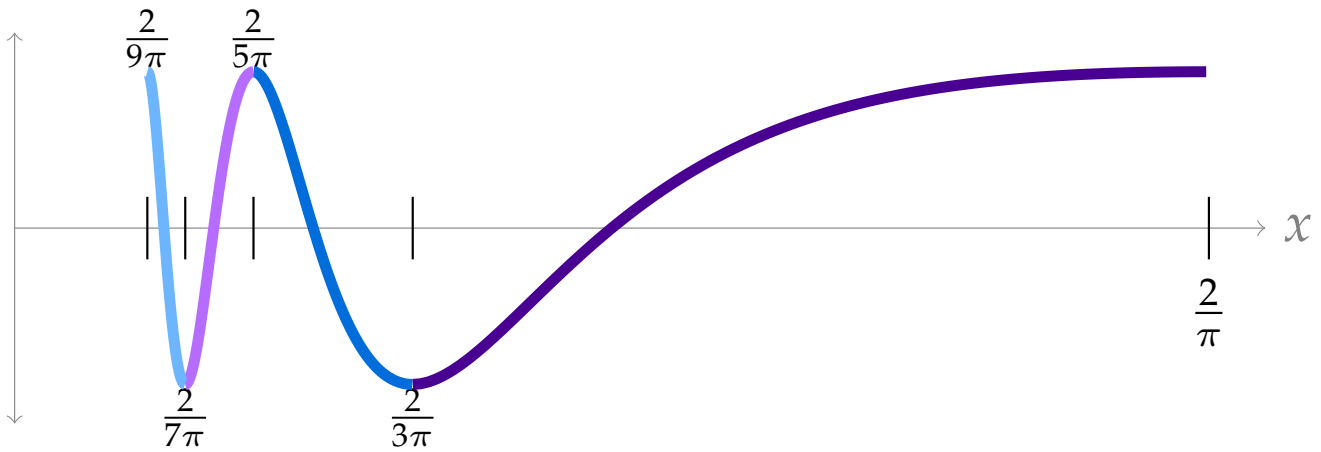
$y = \sin\frac{1}{x}$



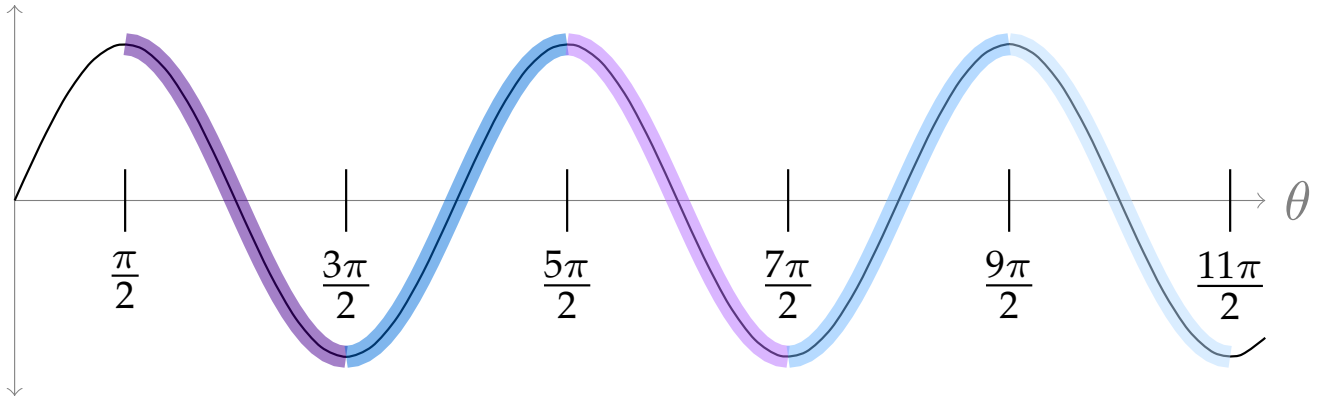
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



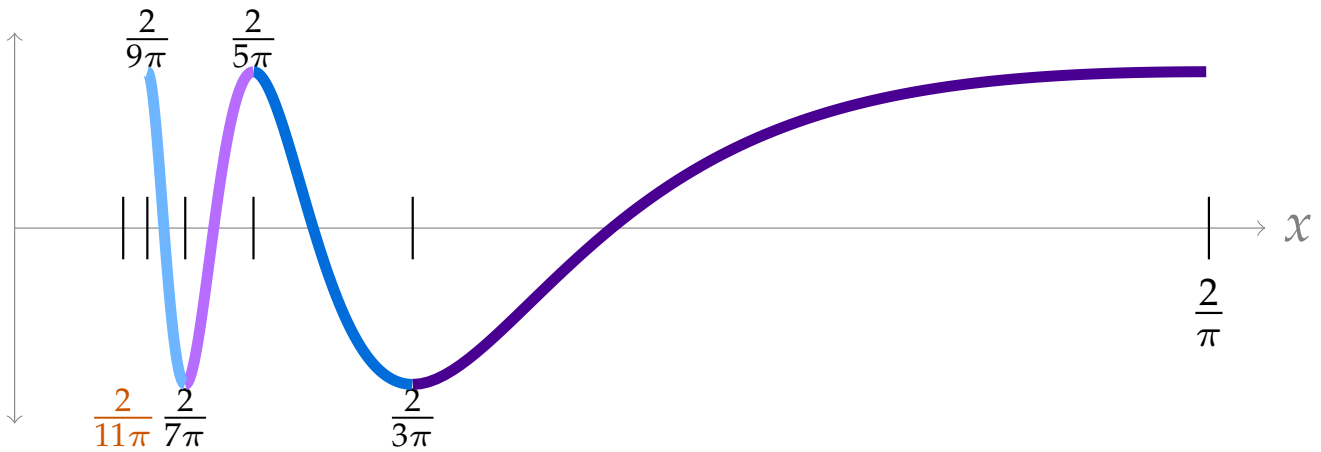
$y = \sin\frac{1}{x}$



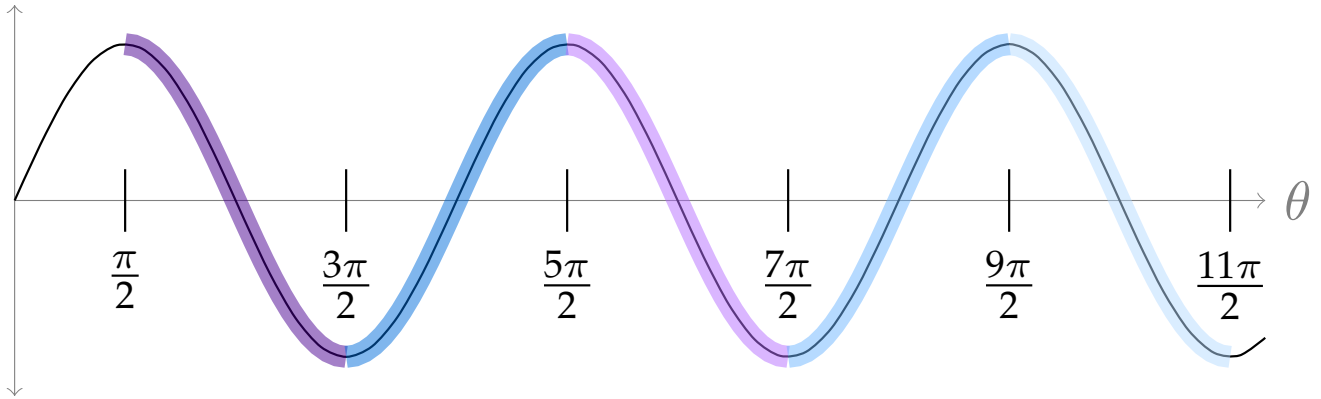
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



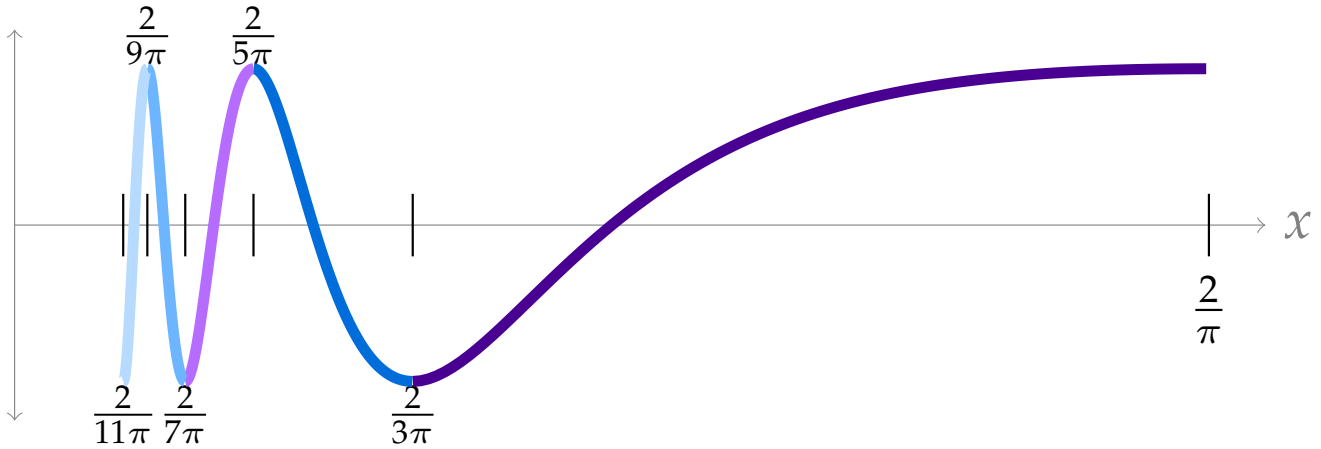
$y = \sin\frac{1}{x}$



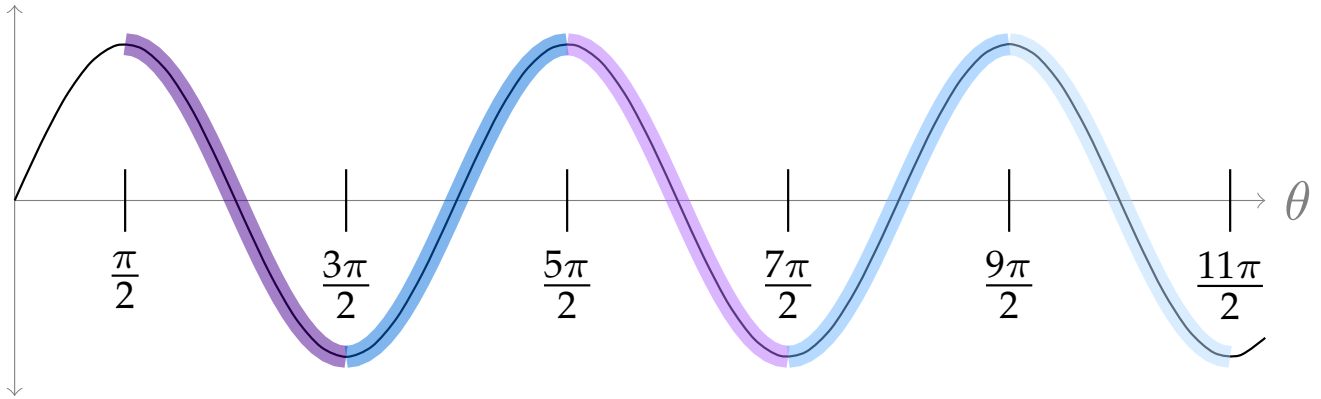
$y = \sin\theta$

Optional: Sketching $f(x) = \sin\left(\frac{1}{x}\right)$

SKIP SKETCHING



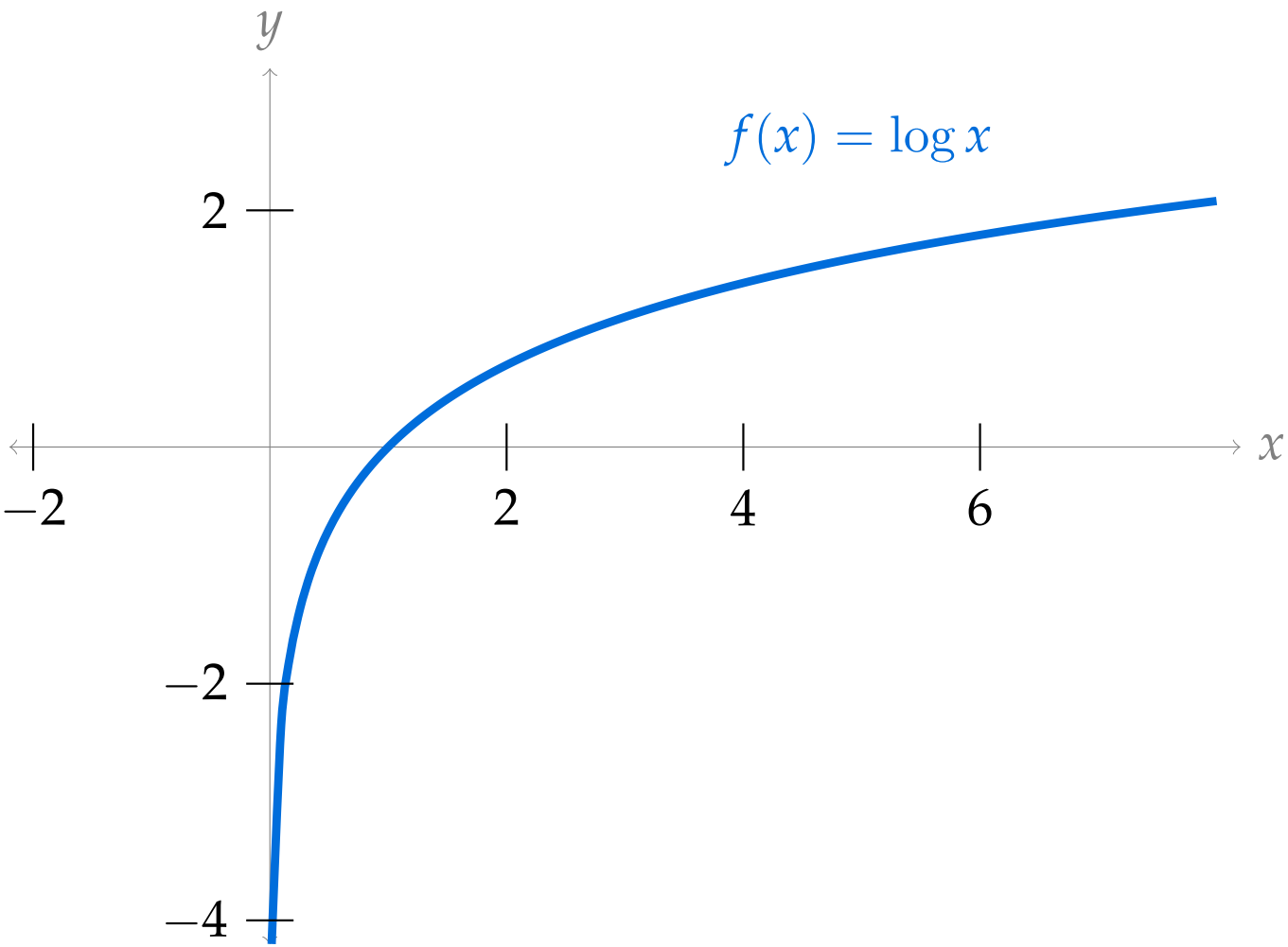
$y = \sin\frac{1}{x}$



$y = \sin\theta$

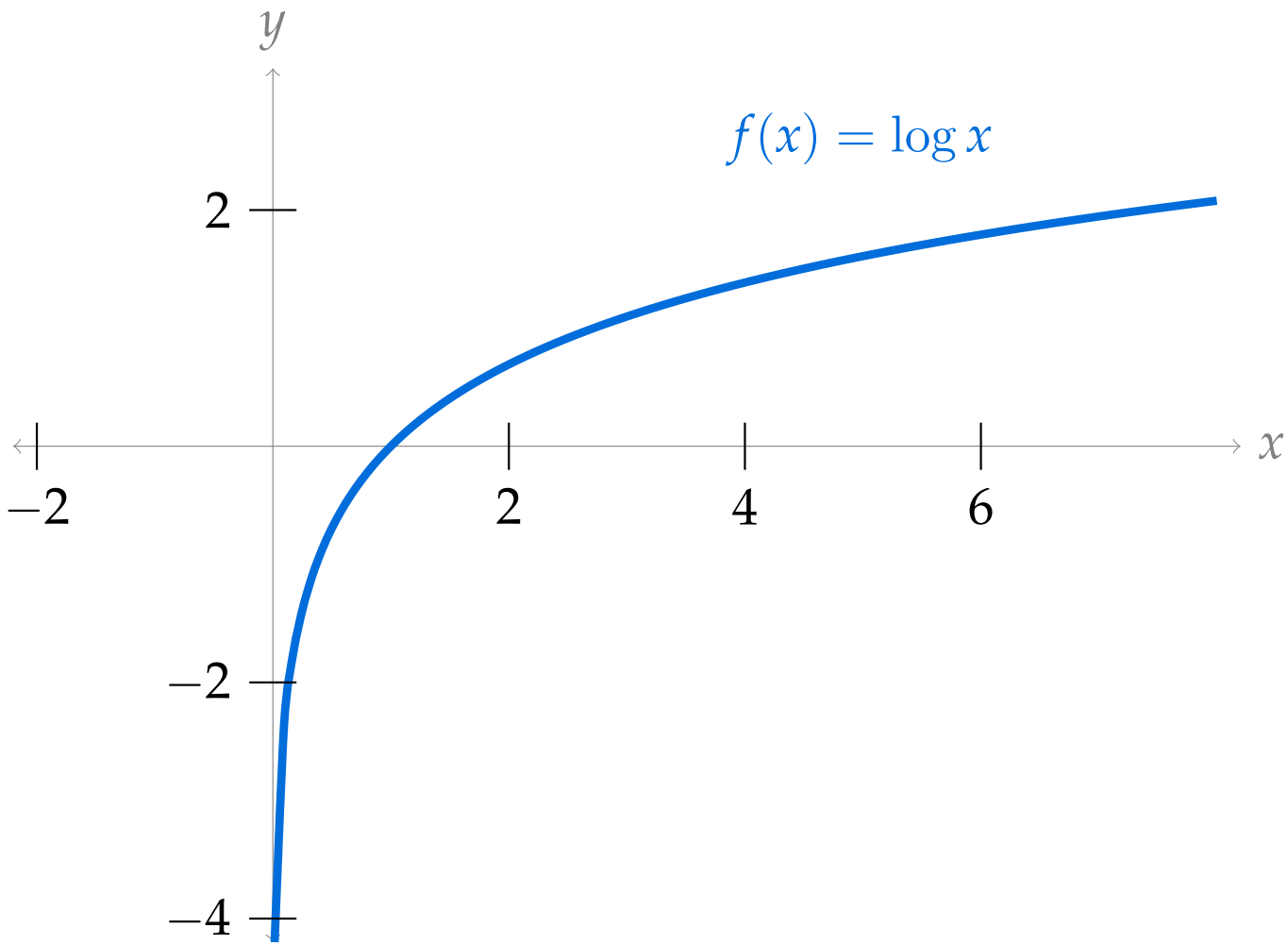
Limits and the Natural Logarithm

Where is $f(x)$ defined, and where is it not defined?



Limits and the Natural Logarithm

What can you say about the limit of $f(x)$ near 0?

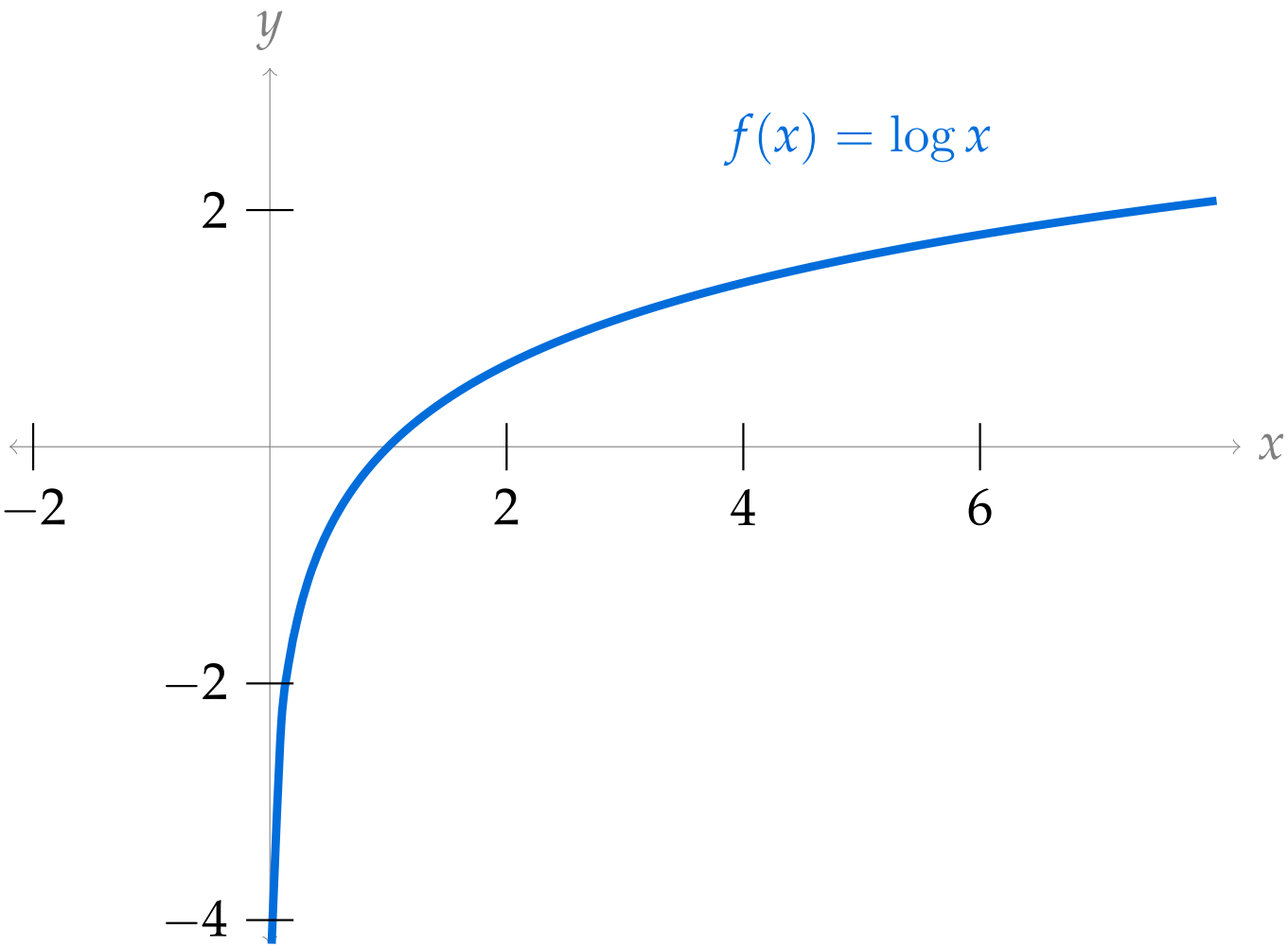


Q

ans

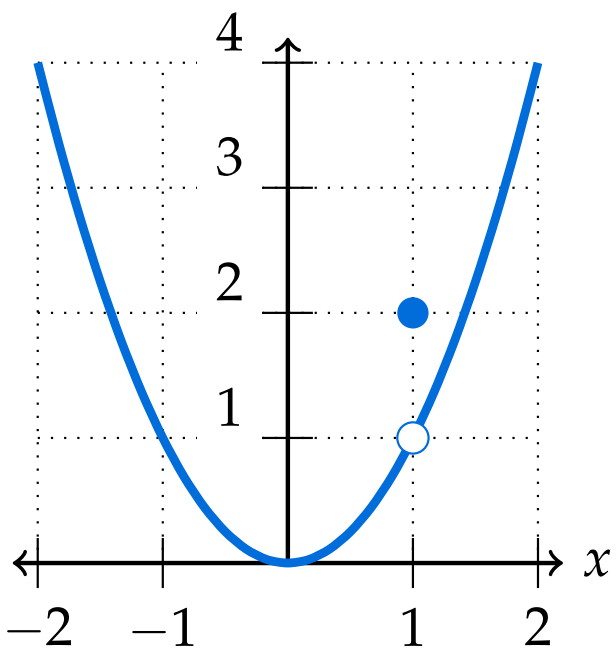
Limits and the Natural Logarithm

What can you say about the limit of $f(x)$ near 0? $\lim_{x \to 0^+} \log(x) = -\infty$



Section 1.3 Review

$$f(x) = \begin{cases} x^2 & x \neq 1 \\ 2 & x = 1 \end{cases}$$



What is $\lim_{x \to 1} f(x)$?

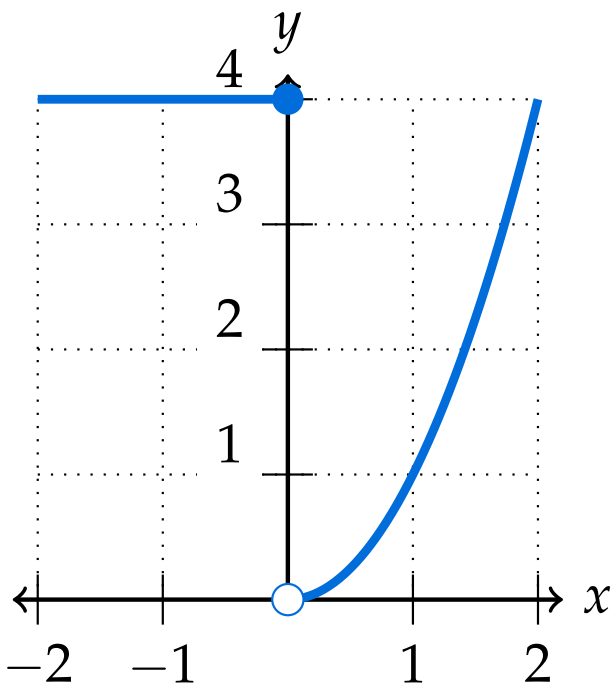
A. $\lim_{x \to 1} f(x) = 2$

B. $\lim_{x \to 1} f(x) = 1$

C. $\lim_{x \to 1} f(x)$ DNE

D. none of the above

$$f(x) = \begin{cases} 4 & x \le 0 \\ x^2 & x > 0 \end{cases}$$



What is $\lim_{x \to 0} f(x)$?

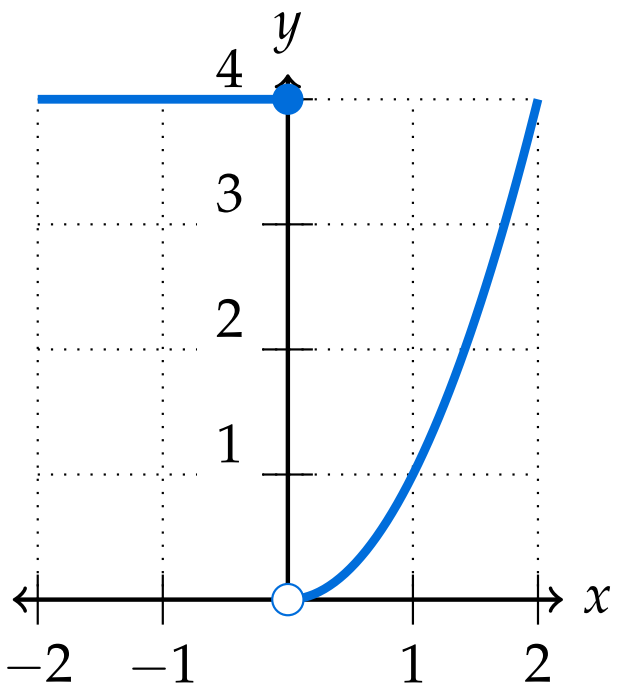
A. $\lim_{x \to 0} f(x) = 4$

B. $\lim_{x \to 0} f(x) = 0$

C. $\lim_{x \to 0} f(x) = \begin{cases} 4 & x \le 0 \\ 0 & x > 0 \end{cases}$

D. none of the above

$$f(x) = \begin{cases} 4 & x \le 0 \\ x^2 & x > 0 \end{cases}$$



What is $\lim_{x \to 0} f(x)$?

A. $\lim_{x \to 0} f(x) = 4$

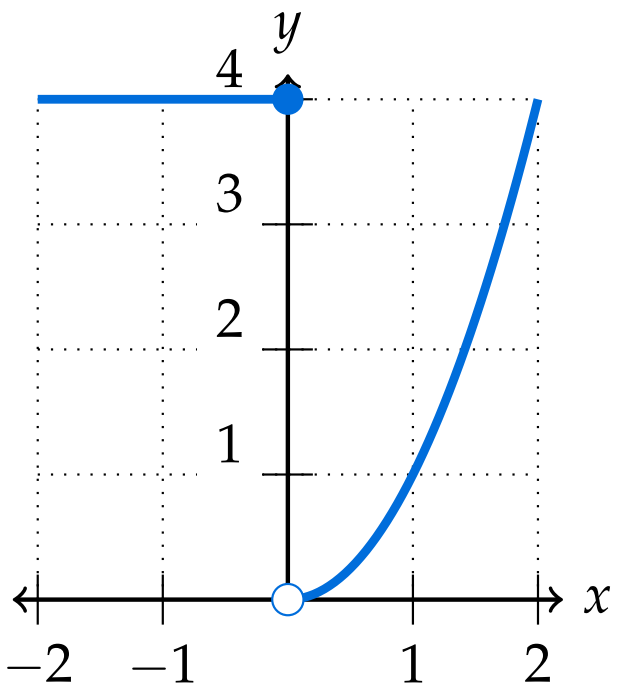
B. $\lim_{x \to 0} f(x) = 0$

C. $\lim_{x \to 0} f(x) = \begin{cases} 4 & x \le 0 \\ 0 & x > 0 \end{cases}$

D. none of the above
$\lim_{x \to 0} f(x)$ DNE

A

$$f(x) = \begin{cases} 4 & x \le 0 \\ x^2 & x > 0 \end{cases}$$



What is $\lim_{x \to 0^+} f(x)$?

A. $\lim_{x \to 0} f(x) = 4$

B. $\lim_{x \to 0} f(x) = 0$

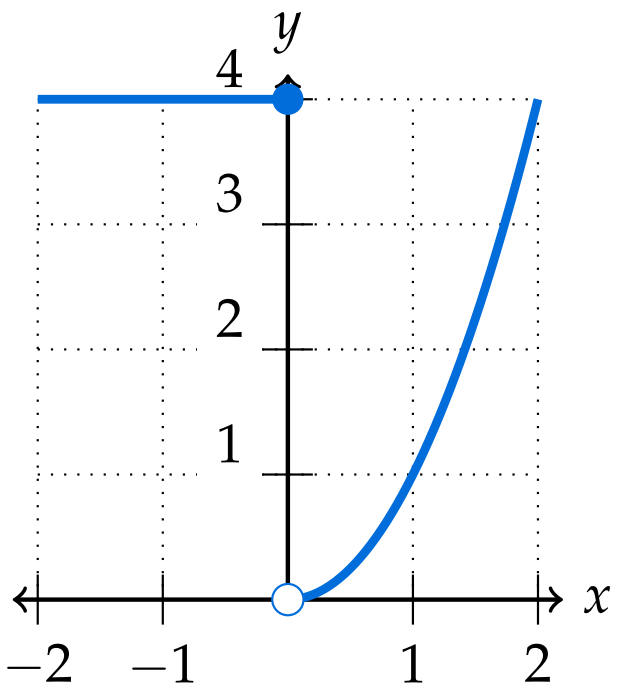
C. $\lim_{x \to 0} f(x) = \begin{cases} 4 & x \le 0 \\ 0 & x > 0 \end{cases}$

D. none of the above

Q

ans

$$f(x) = \begin{cases} 4 & x \le 0 \\ x^2 & x > 0 \end{cases}$$



What is $\lim_{x \to 0^+} f(x)$?

A. $\lim_{x \to 0^+} f(x) = 4$

B. $\lim_{x \to 0^+} f(x) = 0$

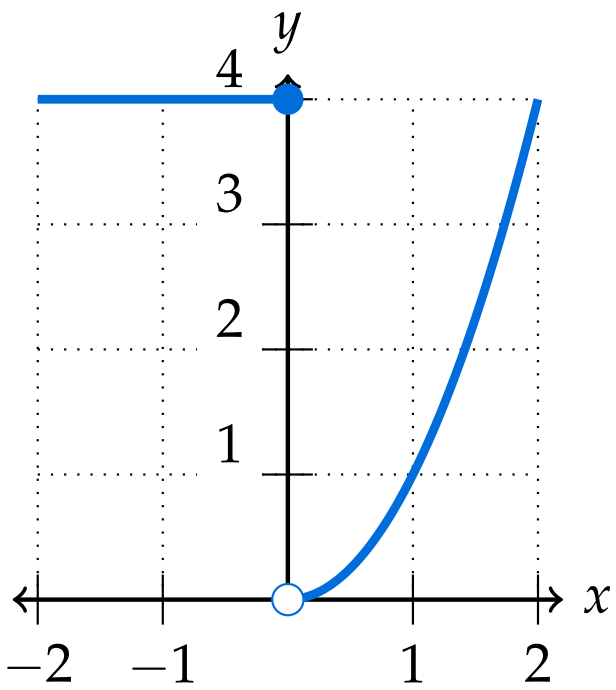
C. $\lim_{x \to 0^+} f(x) = \begin{cases} 4 & x \le 0 \\ 0 & x > 0 \end{cases}$

D. none of the above

A

$$f(x) = \begin{cases} 4 & x \leq 0 \\ x^2 & x > 0 \end{cases}$$

What is $f(0)$?



Suppose $\lim_{x \to 3^-} f(x) = 1$ and $\lim_{x \to 3^+} f(x) = 1.5$.

Does $\lim_{x \to 3} f(x)$ exist?

A. Yes, certainly, because the limits from both sides exist.

B. No, never, because the limit from the left is not the same as the limit from the right.

C. Can't tell. For some functions it might exist, for others not.

Q

ans

Suppose $\lim_{x \to 3^-} f(x) = 1$ and $\lim_{x \to 3^+} f(x) = 1.5$.

Does $\lim_{x \to 3} f(x)$ exist?

A. Yes, certainly, because the limits from both sides exist.

B. No, never, because the limit from the left is not the same as the limit from the right.

C. Can't tell. For some functions it might exist, for others not.

A

Suppose $\lim_{x \to 3^-} f(x) = 22 = \lim_{x \to 3^+} f(x)$.

Does $\lim_{x \to 3} f(x)$ exist?

A. Yes, certainly, because the limits from both sides exist and are equal to each other.

B. No, never, because we only talk about one-sided limits when the actual limit doesn't exist.

C. Can't tell. We need to know the value of the function at $x = 3$.

Q

ans

Suppose $\lim_{x \to 3^-} f(x) = 22 = \lim_{x \to 3^+} f(x)$.

Does $\lim_{x \to 3} f(x)$ exist?

A. Yes, certainly, because the limits from both sides exist and are equal to each other.

B. No, never, because we only talk about one-sided limits when the actual limit doesn't exist.

C. Can't tell. We need to know the value of the function at $x = 3$.

A

Included Work

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